

EEG-Based Emotion Recognition Using Regularized Graph Neural Networks

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Abstract—Electroencephalography (EEG) measures the neuronal activities in different brain regions via electrodes. Many existing studies on EEG-based emotion recognition do not fully exploit the topology of EEG channels. In this article, we propose a regularized graph neural network (RGNN) for EEG-based emotion recognition. RGNN considers the biological topology among different brain regions to capture both local and global relations among different EEG channels. Specifically, we model the inter-channel relations in EEG signals via an adjacency matrix in a graph neural network where the connection and sparseness of the adjacency matrix are inspired by neuroscience theories of human brain organization. In addition, we propose two regularizers, namely node-wise domain adversarial training (NodeDAT) and emotion-aware distribution learning (EmotionDL), to better handle cross-subject EEG variations and noisy labels, respectively. Extensive experiments on two public datasets, SEED, and SEED-IV, demonstrate the superior performance of our model than state-of-the-art models in most experimental settings. Moreover, ablation studies show that the proposed adjacency matrix and two regularizers contribute consistent and significant gain to the performance of our RGNN model. Finally, investigations on the neuronal activities reveal important brain regions and inter-channel relations for EEG-based emotion recognition.

Index Terms—Affective computing, EEG, graph neural network, SEED

1 INTRODUCTION

EMOTION recognition focuses on the recognition of human emotions based on a variety of modalities, such as audio-visual expressions, body language, physiological signals, etc. Compared to other modalities, physiological signals, such as electroencephalography (EEG), electrocardiogram (ECG), electromyography (EMG), etc., have the advantage of being difficult to hide or disguise. In recent years, due to the rapid development of noninvasive, easy-to-use and inexpensive EEG recording devices, EEG-based emotion recognition has received an increasing amount of attention in both research [1] and applications [2].

Emotion models can be broadly categorized into discrete models and dimensional models. The former categorizes emotions into discrete entities, e.g., anger, disgust, fear, happiness, sadness, and surprise in Ekman's theory [3]. The latter describes emotions using their underlying dimensions, e.g., valence, arousal and dominance [4], which measures emotions from unpleasant to pleasant, passive to active, and submissive to dominant, respectively.

EEG signals measure voltage fluctuations from the cortex in the brain and have been shown to reveal important

information about human emotional states [5]. For example, greater relative left frontal EEG activity has been observed when experiencing positive emotions [5]. The voltage fluctuations in different brain regions are measured by electrodes attached to the scalp. Each electrode collects EEG signals in one channel. The collected EEG signals are often analyzed in specific frequency bands, namely delta (1-4 Hz), theta (4-7 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (>30 Hz).

Many existing EEG-based emotion recognition methods are primarily based on the supervised machine learning approach, wherein features are often extracted from preprocessed EEG signals in each channel over a time window. Then, a classifier is trained on the extracted features to recognize emotions. Wang *et al.* [6] compared power spectral density features (PSD), wavelet features and nonlinear dynamical features with a Support Vector Machine (SVM) classifier. Zheng and Lu [7] investigated critical frequency bands and channels using PSD, differential entropy (DE) [8] and PSD asymmetry features, and obtained robust accuracy using deep belief networks (DBN). However, most existing EEG-based emotion recognition approaches do not address the following three challenges: 1) the topological structure of EEG channels are not effectively exploited to learn more discriminative EEG representations; 2) EEG signals vary significantly across different subjects, which hinders the generalizability of the trained classifiers in subject-independent classification settings; and 3) participants may not always generate the intended emotions when watching emotion-eliciting stimuli. Consequently, the emotion labels in the collected EEG data may be noisy and inconsistent with the actual elicited emotions.

There have been several attempts to address the first challenge. Zhang *et al.* [9] and Zhang *et al.* [10] incorporated

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基于脑电图的情绪识别：使用正则化图神经网络

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摘要—脑电图（EEG）通过电极测量不同脑区的神经元活动。许多现有的基于 EEG 的情绪识别研究未能充分利用 EEG 通道的拓扑结构。在本文中，我们提出了一种用于基于 EEG 的情绪识别的正则化图神经网络（RGNN）。RGNN 考虑了不同脑区之间的生物学拓扑结构，以捕捉不同 EEG 通道之间的局部和全局关系。具体而言，我们通过图神经网络中的邻接矩阵对 EEG 信号中的通道间关系进行建模，邻接矩阵的连接性和稀疏性受到人类大脑组织神经科学理论的启发。此外，我们提出了两种正则化器，即节点域对抗训练（NodeDAT）和情绪感知分布学习（EmotionDL），分别用于更好地处理跨被试 EEG 变化和噪声标签。在两个公开数据集 SEED 和 SEED-IV 上的大量实验表明，我们的模型在大多数实验设置中优于现有最佳模型。此外，消融研究表明，所提出的邻接矩阵和两个正则化器对 RGNN 模型的性能提供了持续且显著的提升。最后，对神经元活动的调查揭示了用于基于脑电图的情绪识别的重要脑区和通道间关系。

索引词—情感计算，脑电图，图神经网络，SEED

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1 引言

情绪识别关注基于多种模态（如视听表达、肢体语言、生理信号等）的人体情绪识别。与其他模态相比，生理信号（如脑电图（EEG）、心电图（ECG）、肌电图（EMG）等）具有难以隐藏或伪装的优势。近年来，由于无创、易用且廉价的脑电图记录设备的快速发展，基于脑电图的情绪识别在研究和应用[1][2]中受到了越来越多的关注。情绪模型可以大致分为离散模型和维度模型。前者将情绪分类为离散实体，例如在 Ekman 理论[3]中提到的愤怒、厌恶、恐惧、快乐、悲伤和惊讶。后者则使用其潜在维度来描述情绪，例如效价、唤醒度和支配度[4]，分别从令人不快到愉快、被动到活跃、顺从到支配来衡量情绪。

脑电图（EEG）信号测量大脑皮层电压波动，已被证明能够揭示人类情绪状态的重要信息[5]。例如，在体验积极情绪时，观察到相对左侧额叶 EEG 活动增强[5]。不同脑区的电压波动由头皮上连接的电极测量。每个电极在一个通道中收集 EEG 信号。收集到的 EEG 信号通常在特定频段进行分析，即 delta 波（1-4 Hz）、theta 波（4-7 Hz）、alpha 波（8-13 Hz）、beta 波（13-30 Hz）和 gamma 波（>30 Hz）。

许多现有的基于脑电图（EEG）的情绪识别方法主要基于监督机器学习方法，其中特征通常从每个通道的预处理脑电信号在时间窗口内提取。然后，在提取的特征上训练分类器以识别情绪。Wang 等人[6]比较了功率谱密度特征（PSD）、小波特征和非线性动力学特征与支持向量机（SVM）分类器。Zheng 和 Lu[7]使用 PSD、微分熵（DE）[8]和 PSD 不对称特征研究了关键频段和通道，并使用深度信念网络（DBN）获得了稳健的准确率。然而，大多数现有的基于 EEG 的情绪识别方法未能解决以下三个挑战：1) 未能有效利用 EEG 通道的拓扑结构来学习更具判别性的 EEG 表示；2) EEG 信号在不同受试者之间存在显著差异，这阻碍了训练分类器在受试者无关分类设置中的泛化能力；3) 受试者在观看情绪诱发刺激时可能不会总是产生预期的情绪。因此，收集到的脑电图数据中的情绪标签可能存在噪声，并且与实际诱发的情绪不一致。

已有多种尝试应对第一个挑战。张等人[9]和张等人[10]将

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spatial relations in EEG signals using convolutional neural networks (CNN) and recurrent neural networks (RNN), respectively. However, their approaches require a 2D representation of EEG channels on the scalp, which may cause information loss during flattening because channels are actually arranged in the 3D space. In addition, their approach of using CNNs and RNNs to capture inter-channel relations has difficulty in learning long-range dependencies [11]. Graph neural networks (GNN) has been applied in [12] to capture inter-channel relations using an adjacency matrix. However, similar to CNNs and RNNs, the GNN approach [12] only considers relations between the nearest channels, which thus may lose valuable information between distant channels, such as the PSD asymmetry between channels on the left and right hemispheres in the frontal region, which has been shown to be informative in valence prediction [5].

In recent years, several studies [13], [14] attempted to tackle the second challenge by investigating the transferability of EEG-based emotion recognition models across subjects. Lan *et al.* [15] compared several domain adaptation techniques such as maximum independence domain adaptation (MIDA), transfer component analysis (TCA), subspace alignment (SA), etc. They found that the subject-independent classification accuracy can be improved by around 10 percent. Li *et al.* [16] applied domain adversarial training to lower the influence of individual subject on EEG data and obtained improved performance as well. However, their adversarial training does not exploit any graph structure of the EEG signals and only leads to small performance improvement in our experiment (see Section 7.1).

To the best of our knowledge, no attempt has been made to address the third challenge, i.e., noisy emotion labels, in EEG-based emotion recognition.

In this paper, we propose a regularized graph neural network (RGNN) aiming to address all the three aforementioned challenges. Graph analysis for human brain has been studied extensively in the neuroscience literature [17], [18]. However, making an accurate connectome is still an open question and subject to different scales [18]. Inspired by [12], [19], we consider each EEG channel as a node in our graph. Our RGNN model extends the simple graph convolution network (SGC) [20] and leverages the topological structure of EEG channels. Specifically, we propose a sparse adjacency matrix to capture both local and global inter-channel relations based on the biological topology of human brain [19]. Local inter-channel relations connect nearby groups of neurons and may reveal anatomical connectivity at macroscale [18], [21]. Global inter-channel relations connect distant groups of neurons between the left and right hemispheres and may reveal emotion-related functional connectivity [5], [16].

In addition, we propose a node-wise domain adversarial training (NodeDAT) method to regularize RGNN for better generalization in subject-independent classification scenarios. Different from the domain adversarial training in [16], [22], our NodeDAT method provides a finer-grained regularization by minimizing the domain discrepancies between features in the source and target domains for each channel/node. Moreover, we propose an emotion-aware distribution learning (EmotionDL) method to address the problem of noisy labels in the datasets. Prior studies have shown that noisy labels can adversely impact classification accuracy

[23]. Instead of learning the traditional single-label classification, our EmotionDL method learns a distribution of labels of the training data and thus acts as a regularizer to improve the robustness of our model against noisy labels. Finally, we conduct extensive experiments to validate the effectiveness of our RGNN model and investigate emotion-related informative neuronal activities.

In summary, the main contributions of this paper are as follows:

- 1) We propose a regularized graph neural network (RGNN) model to recognize emotions based on EEG signals. Our biologically inspired model captures both local and global inter-channel relations.
- 2) We propose two regularizers: node-wise domain adversarial training (NodeDAT) and emotion-aware distribution learning (EmotionDL), to improve the robustness of our model against cross-subject variations and noisy labels, respectively.
- 3) We conduct extensive experiments in both subject-dependent and subject-independent classification settings on two public EEG datasets, namely SEED [7] and SEED-IV [24]. Experimental results demonstrate the effectiveness of our proposed model and regularizers. In addition, our RGNN model achieves superior performance over the state-of-the-art models in most experimental settings.
- 4) We investigate the emotional neuronal activities and the results reveal that pre-frontal, parietal and occipital regions may be the most informative regions for emotion recognition. In addition, global inter-channel relations between the left and right hemispheres are important, and local inter-channel relations between (FP1, AF3), (F6, F8) and (FP2, AF4) may also provide useful information.

2 RELATED WORK

In this section, we review related work in the fields of EEG-based emotion recognition, graph neural network, unsupervised domain adaptation, and learning with noisy labels.

2.1 EEG-Based Emotion Recognition

EEG feature extractors and classifiers are the two fundamental components in the machine learning approach of EEG-based emotion recognition. EEG features can be broadly divided into single-channel features and multi-channel ones [25]. The majority of existing features are computed on a single channel, e.g., statistical features [26], PSD [27], differential entropy (DE) [8], and wavelet features [28]. A few number of features are computed on multiple channels to capture the inter-channel relations, e.g., the asymmetry of PSD between two hemispheres [7] and functional connectivity [29], [30], where common indices such as correlation, coherence and phase synchronization were used estimate brain functional connectivity between channels. Another line of research in multi-channel features is to use common spatial filters [31] and spatial-temporal filters [32], [33] to extract class-discriminative EEG features. In contrast, our model is designed to operate on single-channel features and learn to effectively combine them using a graph neural network.

分别使用卷积神经网络（CNN）和循环神经网络（RNN）来提取脑电图（EEG）信号中的空间关系。然而，他们的方法需要将头皮上的 EEG 通道表示为 2D 形式，这在展平过程中可能会导致信息丢失，因为通道实际上是在 3D 空间中排列的。此外，他们使用 CNN 和 RNN 来捕获通道间关系的方法在学习长距离依赖性方面存在困难[11]。图神经网络（GNN）在[12]中已被应用，使用邻接矩阵来捕获通道间关系。然而，与 CNN 和 RNN 类似，GNN 方法[12]仅考虑最近邻通道之间的关系，因此可能会丢失远距离通道之间的宝贵信息，例如额叶区域左右半球通道之间的功率谱密度（PSD）不对称性，这已被证明在效价预测中具有信息价值[5]。

近年来，一些研究[13]、[14]试图通过研究基于脑电图的情绪识别模型在受试者间的可迁移性来应对第二个挑战。Lan 等人[15]比较了几种领域自适应技术，如最大独立领域自适应（MIDA）、迁移成分分析（TCA）、子空间对齐（SA）等。他们发现，受试者无关的分类精度可以提高约 10%。Li 等人[16]应用了领域对抗训练来降低个体受试者对脑电数据的影响，并获得了性能提升。然而，他们的对抗训练并未利用脑电信号的图结构，并且在我们的实验中（见第 7.1 节）仅带来了微小的性能提升。

据我们所知，目前还没有尝试解决基于脑电图的情绪识别中的三个挑战，即噪声情绪标签问题。

在本文中，我们提出了一种正则化图神经网络（RGNN），旨在解决上述三个挑战。对人类大脑的图分析已在神经科学文献中进行了广泛研究[17][18]。然而，构建精确的连接组仍然是一个开放性问题，并且受到不同尺度的影响[18]。受[12][19]的启发，我们将每个脑电图（EEG）通道视为图中的一个节点。我们的 RGNN 模型扩展了简单的图卷积网络（SGC）[20]，并利用了 EEG 通道的拓扑结构。具体而言，我们提出了一种稀疏邻接矩阵，基于人脑的生物拓扑结构[19]，以捕获通道间的局部和全局关系。局部通道间关系连接邻近的神经元群组，并可能揭示宏观尺度的解剖学连接[18][21]。全局通道间关系连接左右半球之间较远的神经元群组，并可能揭示与情绪相关的功能连接[5][16]。

此外，我们提出了一种节点级域对抗训练（NodeDAT）方法来正则化 RGNN，以在主体无关分类场景中实现更好的泛化能力。与[16]、[22]中的域对抗训练不同，我们的 NodeDAT 方法通过最小化源域和目标域中每个通道/节点的特征域差异，提供了一种更细粒度的正则化。此外，我们提出了一种情感感知分布学习（EmotionDL）方法来解决数据集中的噪声标签问题。

先前研究表明，噪声标签会不利地影响分类精度[23]。我们的 EmotionDL 方法不是学习传统的单标签分类，而是学习训练数据的标签分布，从而作为正则化器提高模型对噪声标签的鲁棒性。最后，我们进行了广泛的实验来验证我们的 RGNN 模型的有效性，并研究情感相关的信息性神经元活动。

总之，本文的主要贡献如下：

- 1) 我们提出了一种正则化图神经网络（RGNN）模型，用于基于脑电图（EEG）信号的情绪识别。我们的生物启发模型能够捕捉局部和全局的跨通道关系。
- 2) 我们提出了两种正则化方法：节点级域对抗训练（NodeDAT）和情绪感知分布学习（EmotionDL），分别用于提高模型对跨被试变异和噪声标签的鲁棒性。
- 3) 我们在两个公开的脑电图数据集 SEED [7] 和 SEED-IV [24] 上进行了广泛的实验，包括被试相关和被试无关的分类设置。实验结果表明我们提出的模型和正则化方法的有效性。此外，我们的 RGNN 模型在大多数实验设置中取得了优于现有技术的模型性能。
- 4) 我们研究了情绪相关的神经元活动，结果揭示额叶、顶叶和枕叶区域可能是情绪识别中最具信息量的区域。此外，左右半球之间的全局跨通道关系很重要，而（FP1, AF3）、（F6, F8）和（FP2, AF4）之间的局部跨通道关系也可能提供有用信息。

2 相关工作

在本节中，我们回顾了基于脑电图的情绪识别、图神经网络、无监督域适应和带噪声标签学习领域的相关工作。

2.1 基于脑电图的情绪识别

脑电图特征提取器和分类器是基于脑电图的情绪识别的机器学习方法中的两个基本组件。脑电图特征可以大致分为单通道特征和多通道特征[25]。现有的大多数特征是在单个通道上计算的，例如统计特征[26]、功率谱密度[27]、微分熵（DE）[8]和小波特征[28]。少数特征是在多个通道上计算的，以捕捉通道间的关系，例如两个半球功率谱密度的不对称性[7]以及功能连接性[29]、[30]，其中使用了相关、相干和相位同步等常用指标来估计通道间的脑功能连接性。多通道特征研究的另一条路线是使用公共空间滤波器[31]和时空滤波器[32]、[33]来提取具有类别区分性的脑电图特征。相比之下，我们的模型设计为在单通道特征上运行，并学习使用图神经网络有效地组合它们。

EEG classifiers can be broadly divided into topology-invariant classifiers and topology-aware ones. The majority of existing classifiers are topology-invariant classifiers such as SVM, k-Nearest Neighbors (KNN), DBN [34] and RNN [35], which do not take the topological structure of EEG features into account when learning the EEG representations. In contrast, topology-aware classifiers such as CNN [9], [36], [37] and GNN [12] consider the inter-channel topological relations and learn EEG representations for each channel by aggregating features from nearby channels using convolutional operations either in the euclidean space or in the non-Euclidean space. However, as discussed in Section 1, existing CNNs and GNNs have difficulty in learning the dependencies between distant channels, which may reveal important emotion-related information. Recently, Zhang *et al.* [10] and Li *et al.* [38] proposed to use RNNs to learn spatial topological relations between channels by scanning electrodes in both vertical and horizontal directions. However, their approaches do not fully exploit the topological structure of EEG channels. For example, two topologically close channels may be far away from each other in their scanning sequence. In contrast, our model is able to learn relations between distant channels using global connections.

2.2 Graph Neural Network

Graph neural network (GNN) is a type of neural network dealing with data in the graph domain, e.g., molecular structures, social networks and knowledge graphs [39]. One early work on GNN [40] aimed to learn a converged static state embedding for each node in the graph using a transition function applied to its neighborhood. Later, inspired by the convolutional operation of CNN in the euclidean domain, Bruna *et al.* [41] combined spectral graph theory [42] with neural network and defined convolutional operations in the graph domain using the spectral filters computed from the normalized graph Laplacian. Following this line of research, Defferrard *et al.* [43] proposed fast localized convolutions by using a recursive formulation of the K -order Chebyshev polynomials to approximate the filters. The resulting representation for each node is an aggregation of its K th-order neighborhood. Kipf and Welling [44] further limited $K = 1$ and proposed the standard graph convolutional network (GCN) with a faster localized graph convolutional operation. The convolutional layers in GCN can be stacked K times to effectively convolve the K th-order neighborhood of a node. Recently, Wu *et al.* [20] simplified GCN by removing the non-linearities between convolutional layers in GCN and proposed the simple graph convolution network (SGC), which effectively behaves like a linear feature transformation followed by a logistic regression. Apart from the convolution operation used in GCNs, there are other types of operations used in GNNs, such as attention [45]. However, they are often trained significantly slower than SGC [20]. In this paper, we extend SGC to model EEG signals because it performs orders of magnitude faster than other networks with comparable classification accuracy.

2.3 Unsupervised Domain Adaptation

Unsupervised domain adaptation aims to mitigate the domain shift in knowledge transfer from a supervised source

domain to an unsupervised target domain. The most common approaches are instance re-weighting and domain-invariant feature learning. Instance re-weighting methods [46] aim to infer the resampling weight directly by feature distribution matching across source and target domains in a non-parametric manner. Domain-invariant feature learning methods align features from both source and target domains to a common feature space. The alignment can be achieved by minimizing divergence [47], maximizing reconstruction [48], or adversarial training [22]. Our proposed NodeDAT regularizer extends the domain adversarial training [22] to graph neural networks and achieves finer-grained regularization by minimizing the discrepancies between features in source and target domains for each node individually.

2.4 Learning With Noisy Labels

Commonly adopted approaches to learning with noisy labels are based on the noise transition matrix and robust loss functions. The noise transition matrix specifies the probabilities of transition from each ground truth label to each noisy label and is often applied to modify the cross-entropy loss. The matrix can be pre-computed as *a priori* [49] or estimated from noisy data [50]. A few studies tackle noisy labels by using noise-tolerant robust loss functions, such as unhinged loss [51] and ramp loss [52]. Our proposed EmotionDL regularizer is inspired by [53], which applies distribution learning to classify ambiguous images.

3 PRELIMINARIES

In this section, we introduce the preliminaries of the simple graph convolution network (SGC) [20] and spectral graph convolution, which are the basis of our RGNN model.

3.1 Simple Graph Convolution Network (SGC)

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes a set of nodes and $\mathcal{E}$ denotes a set of edges between nodes in $\mathcal{V}$. Data on $\mathcal{V}$ can be represented by a feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, where $n = |\mathcal{V}|$ and d denotes the input feature dimension. The edge set $\mathcal{E}$ can be represented by a weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with self-loops, i.e., $\mathbf{A}_{ii} = 1, i = 1, 2, \dots, n$. In general, GNNs learn a feature transformation function for $\mathbf{X}$ and produces output $\mathbf{Z} \in \mathbb{R}^{n \times d'}$, where d' denotes the output feature dimension.

Between adjacent layers in GNNs, the feature transformation can be written as

$$\mathbf{H}^{l+1} = f(\mathbf{H}^l, \mathbf{A}), \quad (1)$$

where $l = 0, 1, \dots, L-1$, L denotes the number of layers, $\mathbf{H}^0 = \mathbf{X}$, $\mathbf{H}^L = \mathbf{Z}$, and f denotes the function we want to learn. A simple definition of f would be

$$\mathbf{H}^{l+1} = \sigma(\mathbf{A}\mathbf{H}^l\mathbf{W}^l), \quad (2)$$

where σ denotes a non-linear function and $\mathbf{W}^l$ denotes a weight matrix at layer l . For each node x , function f simply computes the weighted sum of all the node features in its neighborhood including x itself, followed by a non-linear transformation. However, one major limitation of the f in (2) is that repeatedly applying f along multiple layers may

脑电图分类器可以大致分为拓扑不变分类器和拓扑感知分类器。现有的大多数分类器是拓扑不变分类器，如支持向量机（SVM）、k 近邻（KNN）、深度信念网络（DBN）[34]和循环神经网络（RNN）[35]，它们在学习脑电图表示时并未考虑脑电图特征的拓扑结构。相比之下，拓扑感知分类器，如卷积神经网络（CNN）[9]、[36]、[37]和图神经网络（GNN）[12]，考虑了通道间的拓扑关系，并通过在欧几里得空间或非欧几里得空间中通过卷积操作聚合邻近通道的特征来学习每个通道的脑电图表示。然而，正如第 1 节所述，现有的 CNN 和 GNN 在学习远距离通道之间的依赖关系方面存在困难，而这些依赖关系可能揭示重要的情绪相关信息。最近，Zhang 等人[10]和 Li 等人[38]提出使用 RNN 通过在垂直和水平方向扫描电极来学习通道间的空间拓扑关系。然而，他们的方法并未充分利用脑电图通道的拓扑结构。例如，两个拓扑上接近的通道在扫描序列中可能相距很远。相比之下，我们的模型能够通过全局连接学习远距离通道之间的关系。

2.2 图神经网络

图神经网络（GNN）是一种处理图域数据的神经网络，例如分子结构、社交网络和知识图谱[39]。早期关于 GNN 的研究[40]旨在使用应用于其邻域的转换函数为图中的每个节点学习一个收敛的静态状态嵌入。后来，受欧几里得域中 CNN 的卷积操作的启发，Bruna 等人[41]将谱图理论[42]与神经网络相结合，并使用从归一化图拉普拉斯矩阵计算出的谱滤波器在图域中定义了卷积操作。沿着这条研究路线，Defferrard 等人[43]通过使用 K 阶切比雪夫多项式的递归公式来近似滤波器，提出了快速局部卷积。每个节点产生的表示是其 K 阶邻域的聚合。Kipf 和 Welling[44]进一步限制 K=1，并提出了标准图卷积网络（GCN），具有更快的局部图卷积操作。GCN 中的卷积层可以堆叠 K 次，以有效地对节点的 K 阶邻域进行卷积。最近，Wu 等人。[20]通过移除 GCN 中卷积层之间的非线性，简化了 GCN，并提出了简单图卷积网络（SGC），该网络有效地表现为线性特征变换后接逻辑回归。除了 GCN 中使用的卷积操作外，GNN 中还使用其他类型的操作，如注意力[45]。然而，它们通常比 SGC[20]训练速度慢得多。在本文中，我们将 SGC 扩展到建模脑电图（EEG）信号，因为它比其他具有相当分类精度的网络快几个数量级。

2.3 无监督域适应

无监督域适应旨在减轻从监督源知识迁移中的域偏移

将一个有监督的源域映射到一个无监督的目标域。最常见的方法是实例重加权法和域不变特征学习。实例重加权方法[46]旨在通过非参数方式直接根据源域和目标域之间的特征分布匹配来推断重采样权重。域不变特征学习方法将源域和目标域的特征对齐到一个公共特征空间。这种对齐可以通过最小化散度[47]、最大化重建[48]或对抗训练[22]来实现。我们提出的 NodeDAT 正则化器将域对抗训练[22]扩展到图神经网络，并通过最小化每个节点在源域和目标域特征之间的差异来实现更细粒度的正则化。

2.4 基于噪声标签的学习

通常用于处理噪声标签的学习方法基于噪声转移矩阵和鲁棒损失函数。噪声转移矩阵指定了每个真实标签到每个噪声标签的转移概率，通常用于修改交叉熵损失。该矩阵可以预先计算作为先验知识[49]，或从噪声数据中估计[50]。一些研究通过使用容忍噪声的鲁棒损失函数来处理噪声标签，例如 unhinged 损失[51]和 ramp 损失[52]。我们提出的 EmotionDL 正则化器受到[53]的启发，该正则化器将分布学习应用于对模糊图像进行分类。

3 预备知识

在本节中，我们介绍了简单图卷积网络（SGC）[20]和谱图卷积的预备知识，它们是我们 RGNN 模型的基础。

3.1 简单图卷积网络（SGC）

给定一个图 $G = (V, E)$ ，其中 V 表示节点集合， E 表示 V 中节点之间的边集合。 V 上的数据可以用特征矩阵 $X \in \mathbb{R}^{n \times d}$ 表示，其中 $n = |V|$ ， d 表示输入特征维度。边集合 E 可以用带自环的加权邻接矩阵 $A \in \mathbb{R}^{n \times n}$ 表示，即 $A[i, i] = 1$ ， $i = 1, 2, \dots, n$ 。通常，GNNs 学习 X 的特征变换函数，并产生输出 $Z \in \mathbb{R}^{n \times d}$ 。

其中 d 表示输出特征维度。

在 GNNs 的相邻层之间，特征转换可以表示为

$$H = f(H; A); \tag{1}$$

其中 $l = 0; 1; \dots; L-1$ ， L 表示层数， $H = X$ ， $H = Z$ ， f 表示我们想要学习的函数。 f 的一个简单定义是

$$H = s(AH W); \tag{2}$$

其中 s 表示一个非线性函数， W 表示第 l 层的权重矩阵。对于每个节点 x ，函数 f 只是计算其邻域中所有节点特征（包括 x 本身）的加权和，然后进行非线性变换。然而， f 在 (2) 中的一个主要限制是沿多层反复应用 f 可能会

lead to $\mathbf{H}^l$ with overly large values due to summation. Kipf and Welling [44] alleviated this limitation by proposing the graph convolution network (GCN) as follows

$$\mathbf{H}^{l+1} = \sigma(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\mathbf{H}^l\mathbf{W}^l), \quad (3)$$

where $\mathbf{D}$ denotes the diagonal degree matrix of $\mathbf{A}$, i.e., $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$. The normalized adjacency matrix $\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$ prevents $\mathbf{H}$ from growing overly large. If we ignore σ and $\mathbf{W}^l$ temporarily and expand (3), the hidden state $\mathbf{H}_i^{l+1}$ for node $x_i, i = 1, 2, \dots, n$, can be computed via

$$\mathbf{H}_i^{l+1} \leftarrow \frac{\mathbf{A}_{ii}}{\mathbf{D}_{ii} + 1} \mathbf{H}_i^l + \sum_{j=1}^n \frac{\mathbf{A}_{ij}}{\sqrt{(\mathbf{D}_{ii} + 1)(\mathbf{D}_{jj} + 1)}} \mathbf{H}_j^l. \quad (4)$$

Note that each neighboring $\mathbf{H}_j^l$ is now normalized by the degrees of both x_i and x_j . Successively applying L layers aggregates node features within a neighborhood of size L .

To further accelerate training while keeping comparable performance, Wu *et al.* [20] proposed SGC by removing the non-linear function σ in (3) and reparameterizing all linear transformations $\mathbf{W}^l$ across all layers into one linear transformation $\mathbf{W}$ as follows

$$\mathbf{Z} = \mathbf{H}^L = \mathbf{S}\mathbf{H}^{L-1}\mathbf{W}^{L-1} = \dots = \mathbf{S}^L\mathbf{X}\mathbf{W}, \quad (5)$$

where $\mathbf{S} = \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$ and $\mathbf{W} = \mathbf{W}^0\mathbf{W}^1\dots\mathbf{W}^{L-1}$. Essentially, SGC computes a topology-aware linear transformation $\hat{\mathbf{X}} = \mathbf{S}^L\mathbf{X}$, followed by a final linear transformation $\mathbf{Z} = \hat{\mathbf{X}}\mathbf{W}$.

3.2 Spectral Graph Convolution

We analyze GCN from the perspective of spectral graph theory [42]. Graph Fourier analysis relies on the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ or the normalized graph Laplacian $\hat{\mathbf{L}} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$. Since $\hat{\mathbf{L}}$ is a symmetric positive semidefinite matrix, it can be decomposed as $\hat{\mathbf{L}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{U}$ is the orthonormal eigenvector matrix of $\hat{\mathbf{L}}$ and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$ is the diagonal matrix of corresponding eigenvalues. Given graph data $\mathbf{X}$, the graph Fourier transform of $\mathbf{X}$ is $\hat{\mathbf{X}} = \mathbf{U}^T\mathbf{X}$, and the inverse Fourier transform of $\hat{\mathbf{X}}$ is $\mathbf{X} = \mathbf{U}\hat{\mathbf{X}}$. Hence, the graph convolution between $\mathbf{X}$ and a filter $\mathbf{G}$ is computed as follows

$$\mathbf{X} * \mathbf{G} = \mathbf{U}((\mathbf{U}^T\mathbf{G}) \odot (\mathbf{U}^T\mathbf{X})) = \mathbf{U}\hat{\mathbf{G}}\mathbf{U}^T\mathbf{X}, \quad (6)$$

where $\odot$ denotes element-wise multiplication and $\hat{\mathbf{G}} = \text{diag}(\hat{g}_1, \dots, \hat{g}_n)$ denotes a diagonal matrix with n spectral filter coefficients.

To reduce the current learning complexity of $\mathcal{O}(n)$ to that of conventional CNN, i.e., $\mathcal{O}(K)$, (6) can be approximated using the K th order polynomials as follows

$$\mathbf{U}\hat{\mathbf{G}}\mathbf{U}^T\mathbf{X} \approx \mathbf{U}\left(\sum_{i=0}^K \theta_i \mathbf{\Lambda}^i\right)\mathbf{U}^T\mathbf{X} = \sum_{i=0}^K \theta_i \hat{\mathbf{L}}^i \mathbf{X}, \quad (7)$$

where θ_i denotes learnable parameters. To further reduce computational cost, Defferrard *et al.* [43] proposed to use Chebyshev polynomials to approximate the filtering operation as follows

$$\mathbf{U}\hat{\mathbf{G}}\mathbf{U}^T\mathbf{X} = \sum_{i=0}^K \theta_i \mathbf{T}_i(\hat{\mathbf{L}}')\mathbf{X}, \quad (8)$$

where $\hat{\mathbf{L}}' = \frac{2}{\lambda_{\max}}\hat{\mathbf{L}} - \mathbf{I}$ denotes the scaled normalized Laplacian with its eigenvalues lying within $[-1, 1]$, $\lambda_{\max}$ denotes the maximum eigenvalue of $\hat{\mathbf{L}}$, and $\mathbf{T}_i(x)$ denotes the Chebyshev polynomials recursively defined as $\mathbf{T}_i(x) = 2x\mathbf{T}_{i-1}(x) - \mathbf{T}_{i-2}(x)$ with $\mathbf{T}_0(x) = 1$ and $\mathbf{T}_1(x) = x$.

The GCN proposed in [44] made a few approximations to simplify the filtering operation in (8): 1) use $K = 1$; 2) set $\lambda_{\max} = 2$; and 3) set $\theta_1 = -\theta_0$. The resulted GCN arrives at (3). Essentially, the graph convolutional operations defined in (3) and (5) behave like a low-pass filter by smoothing the features of each node on the graph using node features in its neighborhood.

4 REGULARIZED GRAPH NEURAL NETWORK

In this section, we present our regularized graph neural network (RGNN), specifically, the biologically inspired adjacency matrix, the dynamics of RGNN, and two regularizers, i.e., node-wise domain adversarial training (NodeDAT) and emotion-aware distribution learning (EmotionDL).

4.1 Adjacency Matrix in RGNN

The adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ in RGNN represents the topological structure of EEG channels and is essential to graph representation learning, where n denotes the number of channels in EEG signals. Each entry $\mathbf{A}_{ij}$ is learnable and indicates the weight of connection between channels i and j . Note that $\mathbf{A}$ contains self-loops. To reduce overfitting, we model $\mathbf{A}$ as a symmetric matrix by using only $\frac{n(n+1)}{2}$ number of parameters instead of n^2 . Salvador *et al.* [55] observed that the strength of connection between brain regions decays as an inverse square function of physical distance. Hence, we initialize the local inter-channel relations in our adjacency matrix as follows

$$\mathbf{A}_{ij} = \min\left(1, \frac{\delta}{d_{ij}^2}\right), \quad (9)$$

where $d_{ij}, i, j = 1, 2, \dots, n$, denotes the physical distance between channels i and j , which is computed from their 3D coordinates obtained from the data sheet of the recording device, and $\delta > 0$ denotes a calibration constant. Achard and Bullmore [56] observed that sparse fMRI networks, comprising around 20 percent of all possible connections, typically maximize the efficiency of the network topology. Therefore, we choose $\delta = 5$ such that around 20 percent of the entries in $\mathbf{A}$ are non-negligible. We empirically regard entries having values larger than 0.1 as non-negligible connections.

Bullmore and Sporns [19] suggested that the brain organization is shaped by an economic trade-off between minimizing wiring costs and network running costs. Minimizing wiring costs encourages local inter-channel connections as modelled in (9). However, minimizing network running costs encourages certain global inter-channel connections for high efficiency of information transfer across the network as a whole. To this end, we add several global connections to our adjacency matrix to improve the network efficiency. The global connections depend on specific electrode placement

由于求和导致 H 出现过大值。Kipf 和 Welling[44]通过提出图卷积网络（GCN）来缓解这一限制，如下所示

$$H = s(DADHW);$$

(3)

其中 D_p 表示 A 的对角度矩阵，即 $D_p[i,i] = \sum_j A_{ij}$ 。归一化的邻接矩阵 DAD 可以防止 H 过度增长。如果我们暂时忽略 s 和 W ，并展开(3)，节点 x 的隐藏状态 H_i ($i = 1; 2; \dots; n$) 可以通过以下方式计算

$$H_i = \frac{A_{ij}}{D_p[i,i]} H_j = \sum_j \frac{A_{ij}}{D_p[i,i]} H_j$$

(4)

注意，每个相邻的 H_j 现在都通过其邻接矩阵进行归一化。连续应用 L 层将聚合 L 大小邻域内的节点特征。

为了进一步加速训练同时保持相当的性能，Wu 等人[20]提出了 SGC，通过移除(3)中的非线性函数 s 并将所有层的线性变换 W 重新参数化为一个线性变换 W ，如下所示

$$Z = H \frac{1}{D} S H W \frac{1}{D} \dots \frac{1}{D} S X W;$$

(5)

其中 $S = DAD$ ， $W = WW:::W$ 。本质上，SGC 计算一个拓扑感知的线性变换 $X = S X$ ，随后进行一个最终的线性变换 $Z = XW$ 。

3.2 基于谱的图卷积

我们从谱图理论的角度 [42] 分析 GCN。图傅里叶分析依赖于图拉普拉斯矩阵 $L = D - A$ 或归一化图拉普拉斯矩阵 $\hat{L} = I - DAD$ 。由于 $\hat{L}$ 是一个对称正半定矩阵，它可以分解为 $\hat{L} = ULU$ ，其中 U 是 $\hat{L}$ 的正交归一特征向量矩阵， $L = \text{diag}(\lambda_1, \dots, \lambda_n)$ 是相应的特征值对角矩阵。给定图数据 X ， X 的图傅里叶变换是 $\hat{X} = UX$ ，而 $\hat{X}$ 的逆傅里叶变换是 $X = U\hat{X}$ 。因此， X 和滤波器 G 之间的图卷积计算如下

$$X \cdot G = U \delta \delta U G \delta \delta U X \delta \delta = U \hat{G} U X;$$

(6)

其中 δ 表示逐元素乘法， $\hat{G} = \text{diag}(\delta^1; \dots; \delta^n)$ 表示具有 n 个频谱滤波系数的对角矩阵。为了将当前的学习复杂度从 $O(n^2)$ 降低到传统 CNN 的复杂度，即 $O(KP)$ ，(6) 可以使用 K 次多项式近似如下：

$$U \hat{G} U X = U \sum_{i=1}^K u_i L^i U X = \sum_{i=1}^K u_i \hat{L}^i X;$$

(7)

其中 u^d 表示可学习参数。为了进一步降低计算成本，Defferrard 等人[43]提出使用切比雪夫多项式来近似滤波操作，如下所示

$$U \hat{G} U X = \sum_{i=1}^K u_i T_i(\hat{L}) X;$$

(8)

其中 $\hat{L}$ 表示其特征值位于 $[-1, 1]$ 的缩放归一化拉普拉斯矩阵， λ 表示 $\hat{L}$ 的最大特征值， $T_\delta(x)$ 表示递归定义的切比雪夫多项式，定义为 $T_0(x) = 1$ 和 $T_1(x) = x$ 。文献[44]中提出的图卷积网络（GCN）对公式（8）中的滤波操作进行了一些近似简化：1) 使用 $K = 1$ ；2) 设置 $u = 1$ ；3) 设置 $u = u$ 。由此得到的 GCN 得到公式（3）。本质上，公式（3）和（5）中定义的图卷积操作通过使用其邻域节点特征平滑图上每个节点的特征，表现得像一个低通滤波器。

4 正则化图神经网络

在本节中，我们介绍了我们的正则化图神经网络（RGNN），具体包括受生物学启发的邻接矩阵、RGNN 的动态特性以及两个正则化器，即节点级领域对抗训练（NodeDAT）和情感感知分布学习（EmotionDL）。

4.1 RGNN 中的邻接矩阵

RGNN 中的邻接矩阵 $A \in R^{n \times n}$ 表示 EEG 通道的拓扑结构，对于图表示学习至关重要，其中 n 表示 EEG 信号中的通道数量。每个条目 A 是可学习的，表示通道 i 和 j 之间的连接权重。请注意， A 包含自循环。为了减少过拟合，我们通过仅使用 2 个参数而不是 n 来将 A 建模为对称矩阵。Salvador 等人[55]观察到，大脑区域之间的连接强度随着物理距离的平方反比而减弱。因此，我们初始化我们邻接矩阵中的局部通道间关系如下

$$A_{ij} = \min(1, \frac{1}{d_{ij}^2});$$

(9)

其中 $d, i, j = 1, 2, \dots, n$ ，表示通道 i 和 j 之间的物理距离，该距离根据记录设备的数据表获得的 3D 坐标计算得出，且 $d > 0$ 表示校准常数。Achard 和 Bullmore [56] 观察到，稀疏 fMRI 网络（包含所有可能连接中约 20%）通常最大化网络拓扑的效率。因此，我们选择 $d = 5$ ，使得 A 中的约 20% 的条目为非忽略不计的连接。我们经验性地将值大于 0.1 的条目视为非忽略不计的连接。Bullmore 和 Sporns [19] 建议大脑组织是通过最小化布线成本和网络运行成本之间的经济权衡形成的。最小化布线成本鼓励局部通道间连接，如（9）所示。然而，最小化网络运行成本鼓励某些全局通道间连接，以实现整个网络中信息传递的高效率。为此，我们向邻接矩阵中添加一些全局连接以提高网络效率。全局连接取决于特定的电极放置

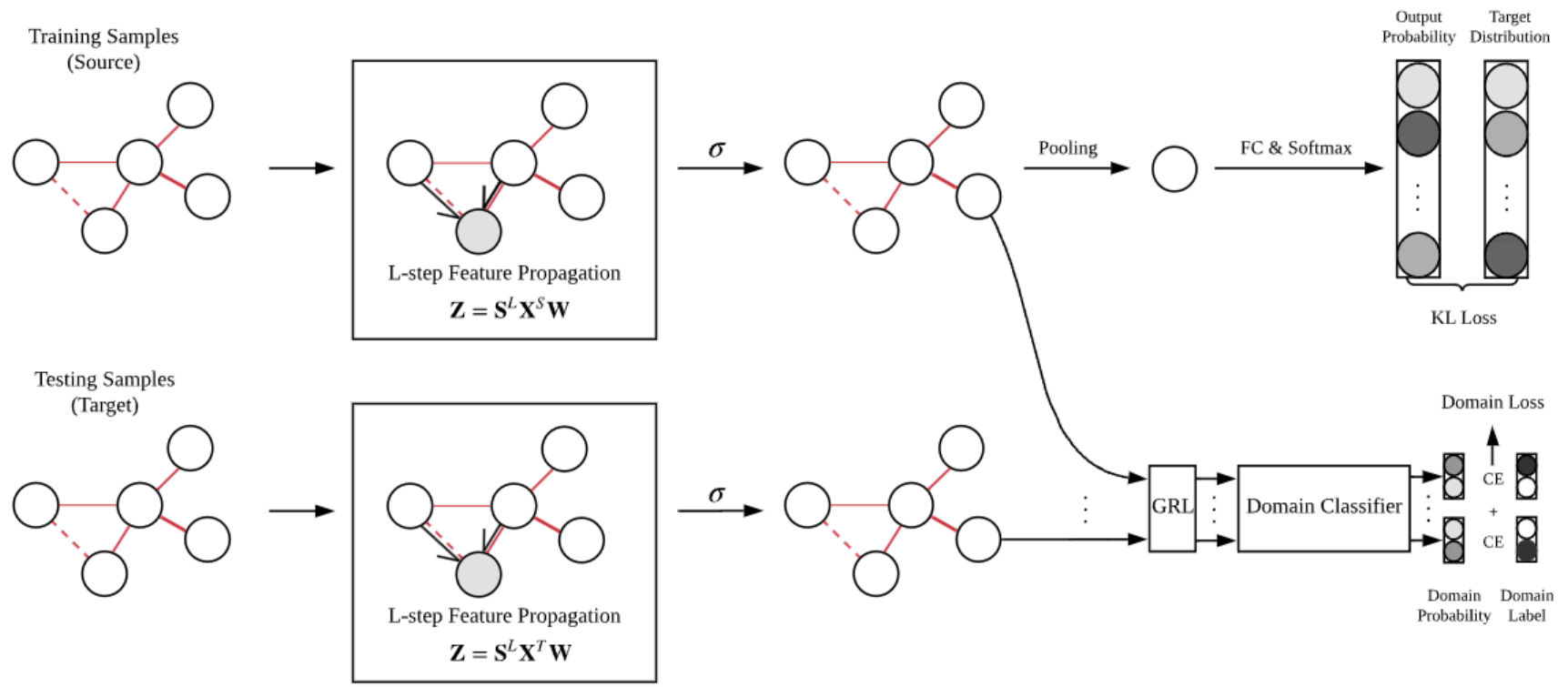


Fig. 1. The overall architecture of our RGNN model. FC denotes fully-connected layer. CE denotes cross-entropy loss. KL denotes Kullback-Leibler divergence [54]. GRL denotes gradient reversal layer [22].

adopted in experiments. Fig. 2 depicts the global connections in both SEED [7] and SEED-IV [24]. The selection of global channels is supported by prior studies showing that the asymmetry in neuronal activities between the left and right hemispheres is informative in valence and arousal predictions [5]. To leverage the differential asymmetry information, we initialize the global inter-channel relations in $\mathbf{A}$ to $[-1, 0]$ as follows

$$\mathbf{A}_{ij} = \mathbf{A}_{ij} - 1, \quad (10)$$

where (i, j) denotes the indices of global channel pairs: (FP1, FP2), (AF3, AF4), (F5, F6), (FC5, FC6), (C5, C6), (CP5, CP6), (P5, P6), (PO5, PO6) and (O1, O2). Note that we select these indices because 1) they are connected to a large number of nodes in their immediate neighborhood, which maximizes the effects of EEG asymmetry; and 2) they empirically perform slightly better than alternative sets of indices (see Section 7.1). Our adjacency matrix $\mathbf{A}$ obtained in (9) and (10) aims to represent the brain network which combines both local anatomical connectivity and emotion-related global functional connectivity.

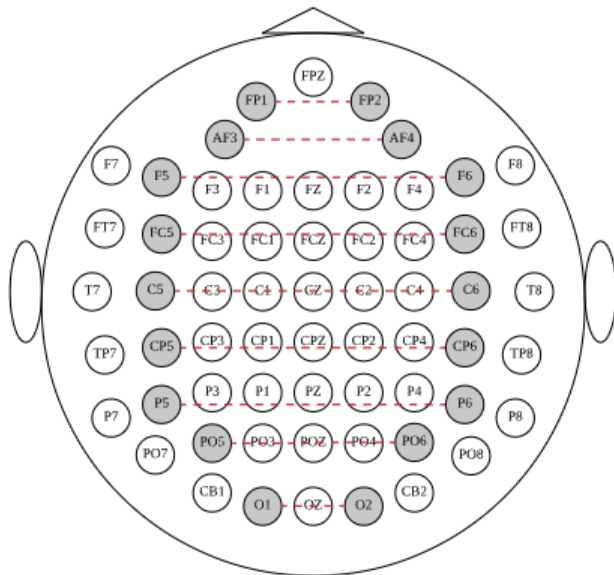


Fig. 2. The 62-channel EEG placement used to collect data in SEED and SEED-IV. Gray symmetric channels are connected globally via red dashed lines.

4.2 Dynamics of RGNN

Our RGNN model extends the SGC model [20]. The architecture of RGNN is illustrated in Fig. 1. Given EEG features $\mathbf{X} \in \mathbb{R}^{N \times n \times d}$ and labels $\mathbf{Y} \in \mathbb{Z}^N$, where N denotes the number of training samples, $\mathbf{Y}_i \in \{0, 1, \dots, C-1\}$ denotes the class index, and C denotes the number of classes. Our model aims to minimize the following cross-entropy loss:

$$\Phi = - \sum_{i=1}^N \log(p(\mathbf{Y}_i | \mathbf{X}_i, \theta)) + \alpha \|\mathbf{A}\|_1, \quad (11)$$

where θ denotes the model parameters we want to optimize, and α denotes the strength of L1 sparse regularization for our adjacency matrix $\mathbf{A}$.

By passing each feature matrix $\mathbf{X}_i$ into our RGNN, the output probability of class $\mathbf{Y}_i$ can be computed as follows

$$\begin{aligned} \mathbf{Z}_i &= \mathbf{S}^L \mathbf{X}_i \mathbf{W}, \\ p(\mathbf{Y}_i | \mathbf{X}_i, \theta) &= \text{softmax}_{\mathbf{Y}_i}(\text{pool}(\sigma(\mathbf{Z}_i)) \mathbf{W}^O), \end{aligned} \quad (12)$$

where $\mathbf{S} \in \mathbb{R}^{n \times n}$, $\mathbf{W} \in \mathbb{R}^{d \times d'}$ and L follow the definitions in (5), $\sigma(x) = \max(0, x)$ denotes a non-linear transformation, $\mathbf{W}^O \in \mathbb{R}^{d' \times C}$ denotes the output weight matrix, and $\text{pool}(\cdot)$ denotes the sum pooling across all nodes on the graph. We choose sum pooling because it demonstrated more expressive power than mean pooling and max pooling [57]. Note that we use the absolute values of $\mathbf{A}$ to compute the degree matrix $\mathbf{D}$ (see (3)) because $\mathbf{A}$ has negative entries, e.g., global connections.

4.2.1 Node-Wise Domain Adversarial Training

EEG signals vary significantly across different subjects, which hinders the generalizability of trained classifiers in subject-independent classification settings. To improve the robustness of our model across subjects, we extend the domain adversarial training [22] by proposing a node-wise domain adversarial training (NodeDAT) method to reduce the discrepancies between source and target domains, i.e., training and testing sets, respectively. Specifically, a domain classifier is proposed to classify each node representation into either

source domain or target domain. During optimization, our model aims to confuse the domain classifier by learning domain-invariant representations. Compared to [22], which only regularizes the pooled representation in the last layer, our NodeDAT method has finer-grained regularization because it explicitly regularizes each node representation before pooling.

Specifically, given labelled source/training data $\mathbf{X}^S \in \mathbb{R}^{N \times n \times d}$ (in this subsection, we denote $\mathbf{X}$ by $\mathbf{X}^S$ for better clarity) and unlabelled target/testing data $\mathbf{X}^T \in \mathbb{R}^{N \times n \times d}$, where in practice $\mathbf{X}^T$ can be either oversampled or downsampled to have the same number of samples as $\mathbf{X}^S$ [22], the domain classifier aims to minimize the sum of the following two binary cross-entropy losses:

$$\Phi_D = - \sum_{i=1}^N \sum_{j=1}^n (\log(p_j(0|\mathbf{X}_i^S, \theta_D)) + \log(p_j(1|\mathbf{X}_i^T, \theta_D))), \quad (13)$$

where θ_D denotes the parameters of the domain classifier, 0 and 1 denote source and target domains, respectively. Intuitively, the domain classifier is learned to classify source data as 0 and target data as 1. The domain probabilities $p_j(\cdot)$ for the j th node on the i th example are computed as

$$\begin{aligned} p_j(0|\mathbf{X}_i^S, \theta_D) &= \text{softmax}_0(\sigma(\mathbf{Z}_{ij}^S) \mathbf{W}^D), \\ p_j(1|\mathbf{X}_i^T, \theta_D) &= \text{softmax}_1(\sigma(\mathbf{Z}_{ij}^T) \mathbf{W}^D), \end{aligned} \quad (14)$$

where $\mathbf{Z}_{ij}^{\{S,T\}}$ denotes the j th node representation in $\mathbf{Z}_i^{\{S,T\}}$, and $\mathbf{W}^D \in \mathbb{R}^{d \times 2}$ denotes the matrix parameter in the domain classifier, i.e., θ_D .

In order to confuse the domain classifier and learn domain invariant node presentation $\mathbf{Z}_{ij}^{\{S,T\}}$, we implement a gradient reversal layer (GRL) [22] that acts like an identity layer in the forward propagation and reverses the gradients of the domain classifier during backpropagation. Consequently, the parameters in the feature extractor essentially perform gradient ascent with respect to the gradients from the domain classifier. The reversed gradients are further scaled by a GRL scaling factor β which gradually increases from 0 to 1 as the training progresses. The gradually increasing β allows our domain classifier to be less sensitive to noisy inputs at the early stages of the training process. Specifically, as suggested in [22], we let $\beta = \frac{2}{1+e^{-10p}} - 1$, where $p \in [0, 1]$ denotes the progression of training.

4.2.2 Emotion-Aware Distribution Learning

Participants may not always generate the intended emotions when watching emotion-eliciting stimuli, which may have negative impact on model performance [23]. To this end, we propose an emotion-aware distribution learning (EmotionDL) method to learn a distribution of classes instead of one single class for each training sample. Specifically, we convert each training label $\mathbf{Y}_i \in \{0, 1, \dots, C-1\}$ into a prior probability distribution of all classes $\hat{\mathbf{Y}}_i \in \mathbb{R}^C$, where $\hat{\mathbf{Y}}_{ic}$ denotes the probability of class c in $\hat{\mathbf{Y}}_i$. The conversion is dataset-dependent. SEED has three classes: negative, neutral, and positive with corresponding class indices 0, 1, and 2, respectively. We convert $\mathbf{Y}$ as follows

$$\hat{\mathbf{Y}}_i = \begin{cases} (1 - \frac{2\epsilon}{3}, \frac{2\epsilon}{3}, 0), & \mathbf{Y}_i = 0, \\ (\frac{\epsilon}{3}, 1 - \frac{2\epsilon}{3}, \frac{\epsilon}{3}), & \mathbf{Y}_i = 1, \\ (0, \frac{2\epsilon}{3}, 1 - \frac{2\epsilon}{3}), & \mathbf{Y}_i = 2, \end{cases} \quad (15)$$

where $\epsilon \in [0, 1]$ denotes a hyper-parameter controlling the noise level in the training labels. This conversion mechanism is based on our assumption that participants are unlikely to generate opposite emotions when watching emotion-eliciting stimuli. Therefore, for each class, the converted class distribution centers on the original class and has zero probabilities at its opposite classes.

SEED-IV has four classes: neutral, sad, fear, and happy with corresponding class indices 0, 1, 2, and 3, respectively. We convert $\mathbf{Y}$ as follows

$$\hat{\mathbf{Y}}_i = \begin{cases} (1 - \frac{3\epsilon}{4}, \frac{\epsilon}{4}, \frac{\epsilon}{4}, \frac{\epsilon}{4}), & \mathbf{Y}_i = 0, \\ (\frac{\epsilon}{3}, 1 - \frac{2\epsilon}{3}, \frac{\epsilon}{3}, 0), & \mathbf{Y}_i = 1, \\ (\frac{\epsilon}{4}, \frac{\epsilon}{4}, 1 - \frac{3\epsilon}{4}, \frac{\epsilon}{4}), & \mathbf{Y}_i = 2, \\ (\frac{\epsilon}{3}, 0, \frac{\epsilon}{3}, 1 - \frac{2\epsilon}{3}), & \mathbf{Y}_i = 3. \end{cases} \quad (16)$$

This conversion is based on the distances between the four emotion classes on the valence-arousal plane. Specifically, in the self-reported ratings [24] for SEED-IV, neutral, sad, fear, and happy movie ratings cluster in the zero valence zero arousal, low valence low arousal, low valence high arousal, and high valence high arousal regions, respectively. We assume that participants are likely to generate emotions that have similar ratings in either valence or arousal dimensions, e.g., both angry and happy have high arousal, but unlikely to generate emotions that are far away in both dimensions, e.g., sad and happy are different in both valence and arousal dimensions.

After obtaining the converted class distributions $\hat{\mathbf{Y}}$, our model can be optimized by minimizing the following Kullback-Leibler (KL) divergence [54] instead of (11):

$$\Phi' = \sum_{i=1}^N \text{KL}(p(\mathbf{Y}|\mathbf{X}_i, \theta), \hat{\mathbf{Y}}_i) + \alpha \|\mathbf{A}\|_1, \quad (17)$$

where $p(\mathbf{Y}|\mathbf{X}_i, \theta)$ denotes the output probability distribution computed via (12). Note that EmotionDL incorporates more prior knowledge than label smoothing, which simply adds uniform noise to other classes.

4.2.3 Optimization of RGNN

Combining both NodeDAT and EmotionDL, the overall loss function Φ'' of RGNN is computed as follows

$$\Phi'' = \Phi' + \Phi_D. \quad (18)$$

The detailed algorithm for training RGNN is presented in Algorithm 1.

5 EXPERIMENTAL SETTINGS

In this section, we present the datasets, classification settings and model settings in our experiments.

5.1 Datasets

We conduct experiments on two public datasets, namely SEED and SEED-IV. The SEED dataset [7] comprises EEG

源域或目标域。在优化过程中，我们的模型旨在通过学习域不变表示来混淆域分类器。与仅对最后一层的池化表示进行正则化的[22]相比，我们的 NodeDAT 方法具有更细粒度的正则化，因为它在池化之前明确地对每个节点表示进行正则化。

具体来说，给定标记的源/训练数据 $X^S \in \mathbb{R}^{N \times d}$ （在本小节中，我们用 X 表示 X^S 以增强清晰度）和未标记的目标/测试数据 $X^T \in \mathbb{R}^{N' \times d}$ ，其中在实践中 X^T 可以是过采样或欠采样以具有与 X^S 相同数量的样本[22]，领域分类器旨在最小化以下两个二元交叉熵损失的乘积：

$$F_{\text{domain}} = \frac{1}{N} \sum_{i=1}^N \log p_{\text{domain}}(y_i | x_i) + \frac{1}{N'} \sum_{j=1}^{N'} \log p_{\text{domain}}(y_j | x_j) \quad (13)$$

其中， θ_{domain} 表示领域分类器的参数，0 和 1 分别表示源域和目标域。直观上，领域分类器被学习为将源数据分类为 0，将目标数据分类为 1。第 i 个样本上第 j 个节点的领域概率 $p_{\text{domain}}(y_j | x_i)$ 计算如下

$$p_{\text{domain}}(y_j | x_i) = \frac{\exp(\theta_{\text{domain}}^T \mathbf{z}_{ij})}{\sum_{k=0,1} \exp(\theta_{\text{domain}}^T \mathbf{z}_{ik})} \quad (14)$$

其中 $\mathbf{z}_{ij}$ 表示 $\mathbf{Z}_i$ 中的第 j 个节点表示， $\mathbf{W}_{\text{domain}} \in \mathbb{R}^{d \times 2}$ 表示领域中的矩阵参数

分类器，即 θ_{domain} 。为了混淆领域分类器并学习领域不变节点表示 $\mathbf{z}_{ij}$ ，我们实现了一个梯度反转层（GRL）[22]，它在前向传播中表现得像一个恒等层，并在反向传播时反转领域分类器的梯度。因此，特征提取器中的参数基本上相对于领域分类器的梯度进行梯度上升。反转的梯度进一步通过一个 GRL 缩放因子 β 进行缩放，该因子随着训练的进行从 0 逐渐增加到 1。逐渐增加的 β 使得我们的领域分类器在训练过程的早期阶段对噪声输入不那么敏感。具体来说，正如[22]中建议的，我们让 $\beta =$

$\frac{1}{1 + \exp(-\beta)}$ ，其中 $\beta \in (0, 1)$ 表示训练的进程。

4.2.2 情感感知分布学习

参与者观看情绪诱发刺激时，可能无法始终产生预期的情绪，这可能会对模型性能产生负面影响[23]。为此，我们提出了一种情绪感知分布学习（EmotionDL）方法，用于学习类别的分布，而不是为每个训练样本学习单个类别。具体来说，我们将每个训练标签 $y \in \{0, 1, \dots, C-1\}$ 转换为所有类别的先验概率分布 $\hat{y} \in \mathbb{R}^C$ ，其中 $\hat{y}_c$ 表示 $\hat{y}$ 中类别 c 的概率。这种转换依赖于数据集。SEED 有三种类别：负面、中性、正面，相应的类别索引分别为 0、1 和 2。我们按如下方式转换 y ：

$$\hat{y} = \begin{bmatrix} \delta_{y=0} \\ \delta_{y=1} \\ \delta_{y=2} \end{bmatrix} \quad (15)$$

其中 $\delta_{y=c}$ 表示一个控制训练标签噪声水平的超参数。这种转换机制基于我们的假设，即参与者在观看情绪诱发刺激时不太可能产生相反的情绪。因此，对于每个类别，转换后的类别分布以原始类别为中心，并在其相反类别处具有零概率。

SEED-IV 有四个类别：中性、悲伤、恐惧和快乐，相应的类别索引分别为 0、1、2 和 3。我们按如下方式转换 y

$$\hat{y} = \begin{bmatrix} \delta_{y=0} \\ \delta_{y=1} \\ \delta_{y=2} \\ \delta_{y=3} \end{bmatrix} \quad (16)$$

这种转换基于四类情绪在效价-唤醒平面上的距离。具体来说，在 SEED-IV 的自我报告评分[24]中，中性、悲伤、恐惧和快乐的电影评分分别聚集在零效价零唤醒、低效价低唤醒、低效价高唤醒和高效价高唤醒区域。我们假设参与者更有可能产生在效价或唤醒维度上具有相似评分的情绪，例如，愤怒和快乐都具有高唤醒，但不太可能产生在两个维度上差异很大的情绪，例如，悲伤和快乐在效价和唤醒维度上都有差异。

在获得转换后的类别分布 $\hat{y}$ 后，我们的模型可以通过最小化以下 Kullback-Leibler (KL) 散度[54]来优化，而不是(11)：

$$F_{\text{KL}} = \frac{1}{N} \sum_{i=1}^N \text{KL}(p_{\text{KL}}(y | x_i) \| p_{\text{KL}}(y | x_i)) \quad (17)$$

其中 $p_{\text{KL}}(y | x_i)$ 表示通过公式(12)计算得到的输出概率分布。需要注意的是，EmotionDL 比标签平滑包含更多的先验知识，而标签平滑只是对其他类别添加均匀噪声。

4.2.3 RGNN 的优化

结合 NodeDAT 和 EmotionDL，RGNN 的整体损失函数 F 计算如下

$$F = F_1 + F_2 \quad (18)$$

RGNN 的训练详细算法在算法 1 中给出。

5 实验设置

在本节中，我们介绍了实验中的数据集、分类设置和模型设置。

5.1 数据集

我们在两个公共数据集上进行了实验，分别是 SEED 和 SEED-IV。SEED 数据集[7]包含脑电图

data of 15 subjects (7 males) recorded in 62 channels using the ESI NeuroScan System.¹ The data were collected when participants watch emotion-eliciting movies in three types of emotions, namely negative, neutral and positive. Each movie lasts around 4 minutes. Three sessions of data are collected and each session comprises 15 trials/movies for each subject. To make a fair comparison with existing studies, we directly use the pre-computed differential entropy (DE) features smoothed by linear dynamic systems (LDS) [7] in SEED. DE extends the idea of Shannon entropy and measures the complexity of a continuous random variable. In SEED, DE features are pre-computed over five frequency bands (delta, theta, alpha, beta and gamma) for each second of EEG signals (without overlapping) in each channel.

Algorithm 1. The Training Algorithm of RGNN

Input: Training samples $\mathbf{X}$ and $\hat{\mathbf{Y}}$, unlabelled testing samples $\mathbf{X}^T$, learning rate η , number of epochs T , batch size B , other regularization hyper-parameters;

Output: The learned model parameters in RGNN;

- 1: Randomly initialize model parameters in RGNN using Xavier initialization [58];
 - 2: Initialize adjacency matrix $\mathbf{A}$ based on (9) and (10);
 - 3: **for** $i = 1: T$ **do**
 - 4: **repeat**
 - 5: Draw one batch of training samples $\mathbf{X}_B$ and $\hat{\mathbf{Y}}_B$ from $\mathbf{X}$ and $\hat{\mathbf{Y}}$, respectively;
 - 6: Draw one batch of testing samples $\mathbf{X}_B^T$ from $\mathbf{X}^T$;
 - 7: Compute degree matrix $\mathbf{D}$ based on (3);
 - 8: Compute normalized adjacency matrix $\mathbf{S}$ based on (5);
 - 9: Compute output representation $\mathbf{Z}$ based on (12);
 - 10: Use $\mathbf{X}_B$ and $\hat{\mathbf{Y}}_B$ to compute KL loss Φ' based on (17);
 - 11: Use $\mathbf{X}_B$ and $\mathbf{X}_B^T$ to compute domain loss Φ_D based on (13);
 - 12: Compute GRL scaling factor β ;
 - 13: Update $\mathbf{W}^D \leftarrow \mathbf{W}^D - \eta \frac{\partial \Phi_D}{\partial \mathbf{W}^D}$;
 - 14: Update $\mathbf{W}^O \leftarrow \mathbf{W}^O - \eta \frac{\partial \Phi'}{\partial \mathbf{W}^O}$;
 - 15: Update $\mathbf{W} \leftarrow \mathbf{W} - \eta (\frac{\partial \Phi'}{\partial \mathbf{W}} - \beta \frac{\partial \Phi_D}{\partial \mathbf{W}})$;
 - 16: Update $\mathbf{A} \leftarrow \mathbf{A} - \eta (\frac{\partial \Phi'}{\partial \mathbf{A}} - \beta \frac{\partial \Phi_D}{\partial \mathbf{A}})$;
 - 17: **until** all samples in $\mathbf{X}$ have been drawn;
-

The SEED-IV dataset [24] comprises EEG data of 15 subjects (7 males) recorded in 62 channels.² The recording device is the same as the one used in SEED. The data were collected when participants watch emotion-eliciting movies in four types of emotions, namely neutral, sad, fear, and happy. Each movie lasts around 2 minutes. Three sessions of data are collected and each session comprises 24 trials/movies for each subject. Similar to SEED, we adopt the pre-computed DE features from SEED-IV.

5.2 Classification Settings

We closely follow prior studies to conduct both subject-dependent and subject-independent classifications on both SEED and SEED-IV to evaluate our model.

5.2.1 Subject-Dependent Classification

For SEED, we follow the experimental settings in [7], [12], [16] to evaluate our RGNN model using subject-dependent classification. Specifically, for each subject, we train our model using the first 9 trials as the training set and the remaining 6 trials as the testing set. We evaluate the model performance by using the accuracy averaged across all subjects over two sessions of EEG data [7]. Similarly, for subject-dependent classification on SEED-IV, we follow the experimental settings in [24], [38] to use the first 16 trials for training and the remaining 8 trials containing all emotions (two trials per emotion class) for testing. We evaluate our model using data from all three sessions [24].

5.2.2 Subject-Independent Classification

For SEED, we follow the experimental settings in [12], [13], [16] to evaluate our RGNN model using subject-independent classification. Specifically, we adopt leave-one-subject-out cross-validation, i.e., during each fold, we train our model on 14 subjects and test on the remaining subject. We evaluate the model performance using the accuracy averaged across all test subjects over one session of EEG data [13]. Similarly, for SEED-IV, we follow the experimental settings in [38] to evaluate our RGNN model using subject-independent classification. We evaluate our model using data from all three sessions [38].

5.3 Model Settings in RGNN

For hyper-parameters of RGNN in all experiments, we empirically set the number of convolutional layers $L = 2$, dropout rate of 0.7 at the output fully-connected layer [64], and batch size of 16. We use Adam [65] to optimize model parameters using gradient descent. We only tune the output feature dimension d' , label noise level ϵ , learning rate η , L1 regularization factor α , and L2 regularization for each experiment. Note that we only adopt NodeDAT in subject-independent classification experiments. Our model is publicly available.³ We compare our model with several baselines, which are all cited from published results [10], [12], [16], [38].

6 PERFORMANCE EVALUATIONS

In this section, we present model evaluation results and investigate the critical frequency bands and confusion matrices of our RGNN model.

6.1 Subject-Dependent Classification

Table 1 presents the subject-dependent classification accuracy of our RGNN model and all baselines on both SEED and SEED-IV. The performance on SEED in the individual delta, theta, alpha, beta, and gamma bands is reported as well. It is encouraging to see that our model achieves better performance than all baselines including the state-of-the-art BiHDM on both datasets when features from all frequency bands are used. In particular, our model performs better than DGCNN, another GNN-based model that leverages the topological structure of EEG channels. Besides the proposed two regularizers (see Table 3), the main performance

1. <https://compumedicsneuroscan.com/>

2. SEED-IV also contains eye movement data, which we do not use in our experiments.

3. <https://github.com/zhongpeixiang/RGNN>

使用 ESI NeuroScan 系统记录了 15 名受试者（7 名男性）在 62 个通道上的数据。数据收集时，受试者观看引发情绪的电影，情绪类型分为负面、中性及正面三种。每部电影的时长约为 4 分钟。共收集了三次数据，每次数据包含每位受试者的 15 次试验/电影。为了与现有研究进行公平比较，我们直接使用了 SEED 中预先计算并经过线性动态系统（LDS）平滑的微分熵（DE）特征[7]。DE 扩展了香农熵的思想，用于测量连续随机变量的复杂性。在 SEED 中，DE 特征预先计算了每个通道中每秒 EEG 信号（无重叠）在五个频段（delta、theta、alpha、beta 和 gamma）上的值。

算法 1. RGNN 的训练算法输入：训练样本 X 和 $\hat{Y}$ ，无标签测试样本 X ，学习率 h ，训练轮数 T ，批次大小 B ，其他正则化超参数；输出：RGNN 中学习到的模型参数；1：使用 Xavier 初始化 [58] 随机初始化 RGNN 中的模型参数；2：根据 (9) 和 (10) 初始化邻接矩阵 A ；

3: for $i = 1: T$ do
4: repeat
5: 从 X 和 $\hat{Y}$ 中分别抽取一批训练样本 X 和 $\hat{Y}$ ；6: 从 X 中抽取一批测试样本 X ；7: 根据 (3) 计算度矩阵 D ；8: 根据 (5) 计算归一化邻接矩阵 S ；9: 根据 (12) 计算输出表示 Z ；10: 使用 X 和 $\hat{Y}$ 计算 KL 损失 F

基于(17)；11: 使用 X 和 X 计算域损失 F 基于(13)；12: 计算 GRL 缩放因子 b ；13: 更新 $W W h @$ ；14: 更新 $W W h$

$$15: \text{更新 } W \quad W h \tilde{\partial} \quad @W b @P;$$
$$16: \text{更新 } A \quad A h \tilde{\partial} \quad @A b @P;$$

17: 直到 X 中的所有样本都被抽取；

SEED-IV 数据集[24]包含 15 名受试者（7 名男性）在 62 个通道上记录的脑电图数据。记录设备与 SEED 中使用的相同。数据是在受试者观看引发情绪的电影时收集的，这些电影分为四种情绪类型：中性、悲伤、恐惧和快乐。每部电影持续约 2 分钟。收集了三次数据，每次包含每个受试者的 24 次试验/电影。与 SEED 类似，我们采用了从 SEED-IV 预计算的 DE 特征。

5.2 分类设置

我们紧密遵循先前研究，在 SEED 和 SEED-IV 上同时进行受试者相关和受试者无关的分类，以评估我们的模型。

5.2.1 受试者相关分类

对于 SEED，我们遵循[7]、[12]、[16]中的实验设置，使用受试者相关分类来评估我们的 RGNN 模型。具体来说，对于每个受试者，我们使用前 9 次试验作为训练集，剩余 6 次试验作为测试集来训练模型。我们通过在两次 EEG 数据会话中所有受试者的平均准确率来评估模型性能[7]。类似地，对于 SEED-IV 的受试者相关分类，我们遵循[24]、[38]中的实验设置，使用前 16 次试验进行训练，剩余 8 次试验（包含所有情绪，每种情绪类别两次试验）进行测试。我们使用来自三个会话的数据来评估我们的模型[24]。

5.2.2 受试者无关分类

对于 SEED，我们遵循[12]、[13]、[16]中的实验设置，使用受试者无关分类来评估我们的 RGNN 模型。具体来说，我们采用留一受试者交叉验证，即在每个折中，我们在 14 个受试者上训练模型，并在剩余的受试者上测试。我们使用在单个脑电数据会话中所有测试受试者上的平均准确率来评估模型性能[13]。类似地，对于 SEED-IV，我们遵循[38]中的实验设置，使用受试者无关分类来评估我们的 RGNN 模型。我们使用来自所有三个会话的数据来评估我们的模型[38]。

5.3 RGNN 中的模型设置

在所有实验中，RGNN 的超参数我们通过经验设置为卷积层数 $L=2$ ，输出全连接层的 dropout 率为 0.7[64]，以及批大小为 16。我们使用 Adam[65]通过梯度下降优化模型参数。我们仅调整输出特征维度 d 、标签噪声水平 ϵ 、学习率 h 、L1 正则化因子 a 以及 L2 正则化。请注意，在受试者无关分类实验中，我们仅采用 NodeDAT。我们的模型是公开可用的。我们将我们的模型与几个基线进行了比较，这些基线均来自已发表成果[10]、[12]、[16]、[38]。

6 性能评估

在本节中，我们展示模型评估结果，并研究我们 RGNN 模型的关键频段和混淆矩阵。

6.1 受试者相关分类

表 1 展示了我们的 RGNN 模型和所有基线在 SEED 和 SEED-IV 上的受试者相关分类准确率。同时报告了在 SEED 上各个频率带（delta、theta、alpha、beta 和 gamma）的性能。令人鼓舞的是，当使用所有频率带特征时，我们的模型在两个数据集上均优于所有基线，包括最先进的 BiHDM。特别是，我们的模型性能优于 DGCNN，这是一种利用脑电图通道拓扑结构的基于 GNN 的模型。除了提出的两种正则化器（见表 3）外，主要性能

1. <https://compumedicsneuroscan.com/SEED-IV> 还包含眼动数据，但我们未在实验中使用。

3. <https://github.com/zhongpeixiang/RGNN>

TABLE 1
Subject-Dependent Classification Accuracy (Mean/Std) on SEED and SEED-IV

Model	SEED						SEED-IV
	delta band	theta band	alpha band	beta band	gamma band	all bands	all bands
SVM	60.50/14.14	60.95/10.20	66.64/14.41	80.76/11.56	79.56/11.38	83.99/09.92	56.61/20.05
GSCCA [59]	63.92/11.16	64.64/10.33	70.10/14.76	76.93/11.00	77.98/10.72	82.96/09.95	69.08/16.66
DBN [7]	64.32/12.45	60.77/10.42	64.01/15.97	78.92/12.48	79.19/14.58	86.08/08.34	66.77/07.38
STRNN [10]	80.90 /12.27	83.35 /09.15	82.69 /12.99	83.41/10.16	69.61/15.65	89.50/07.63	-
DGCNN [12]	74.25/11.42	71.52/05.99	74.43/12.16	83.65/10.17	85.73/10.64	90.40/08.49	69.88/16.29
BiDANN [16]	76.97/10.95	75.56/07.88	81.03/11.74	89.65 /09.59	88.64/09.46	92.38/07.04	70.29/12.63
EmotionMeter [24]	-	-	-	-	-	-	70.58/17.01
BiHDM [38] (SOTA)	-	-	-	-	-	93.12/06.06	74.35/14.09
RGNN (Our model)	76.17/07.91	72.26/07.25	75.33/08.85	84.25/12.54	89.23 /08.90	94.24 /05.95	79.37 /10.54

TABLE 2
Subject-Independent Classification Accuracy (Mean/Std) on SEED and SEED-IV

Model	SEED						SEED-IV
	delta band	theta band	alpha band	beta band	gamma band	all bands	all bands
SVM	43.06/08.27	40.07/06.50	43.97/10.89	48.63/10.29	51.59/11.83	56.73/16.29	37.99/12.52
TCA [60]	44.10/08.22	41.26/09.21	42.93/14.33	43.93/10.06	48.43/09.73	63.64/14.88	56.56/13.77
SA [61]	53.23/07.47	50.60/08.31	55.06/10.60	56.72/10.78	64.47/14.96	69.00/10.89	64.44/09.46
T-SVM [62]	-	-	-	-	-	72.53/14.00	-
DGCNN [12]	49.79/10.94	46.36/12.06	48.29/12.28	56.15/14.01	54.87/17.53	79.95/09.02	52.82/09.23
DAN [63]	-	-	-	-	-	83.81/08.56	58.87/08.13
BiDANN-S [16]	63.01/07.49	63.22 /07.52	63.50 /09.50	73.59/09.12	73.72/08.67	84.14/06.87	65.59/10.39
BiHDM [38] (SOTA)	-	-	-	-	-	85.40 /07.53	69.03/08.66
RGNN (Our model)	64.88 /06.87	60.69/05.79	60.84/07.57	74.96 /08.94	77.50 /08.10	85.30/06.72	73.84 /08.02

improvement can be attributed to two factors: 1) our adjacency matrix incorporates the emotion-discriminative global inter-channel asymmetry relation between the left and right hemispheres; and 2) our model has less concern of overfitting by extending SGC, which is much simpler than ChebNet [43] used in DGCNN.

6.2 Subject-Independent Classification

Similar to Tables 1 and 2 presents the subject-independent classification results. When using features from all frequency bands, our model performs marginally worse than BiHDM on SEED but much better than BiHDM on SEED-IV (nearly 5 percent improvement). In addition, our model achieves the lowest standard deviation in accuracy compared to all

baselines on both datasets, showing the robustness of our model against cross-subject variations.

Comparing the results shown in Tables 1 and 2, we find that the accuracy obtained in subject-independent settings is consistently worse than the accuracy obtained in subject-dependent settings by around 5 to 30 percent for every model. This finding is unsurprising because the variability of EEG signals across subjects makes subject-independent classification more challenging. However, an interesting observation is that the performance gap between these two settings is gradually decreasing from around 27 percent on SEED and 19 percent on SEED-IV using SVM to around 9 percent on SEED and 6 percent on SEED-IV using our model. One possible reason for the diminishing performance gap is that recent deep learning models in subject-independent classification settings are becoming better at leveraging a large amount of data and learning subject-invariant EEG representations. This observation seems to indicate that transfer learning may be a necessary tool for emotion recognition in cross-subject settings.

6.3 Performance Comparison of Frequency Bands

We further compare the performance of our model and all baselines on SEED using features from different frequency bands, as reported in Tables 1 and 2. In subject-dependent experiments, STRNN achieves the highest accuracy in delta, theta and alpha bands, BiDANN performs best in beta band, and our model performs best in gamma band. In subject-independent experiments, BiDANN-S achieves the highest accuracy in theta and alpha bands, and our model performs best in delta, beta and gamma bands.

TABLE 3
Ablation Study for Subject-Independent Classification Accuracy (Mean/Std) on SEED and SEED-IV

Model	SEED	SEED-IV
RGNN	85.30 /06.72	73.84 /08.02
correlation-based adjacency matrix	84.41/06.94	72.73/08.36
coherence-based adjacency matrix	84.02/07.05	72.26/08.48
random adjacency matrix	83.57/07.34	71.78/08.64
– symmetric adjacency matrix	83.69/07.92	72.02/08.66
– global connection	82.42/08.24	71.13/08.78
global connection alternative 1	84.52/06.87	73.29/08.18
global connection alternative 2	84.23/07.04	73.08/08.35
– NodeDAT	81.92/09.35	71.65/09.43
DAT	83.51/08.11	72.40/08.54
– EmotionDL	82.27/08.81	70.76/09.22

Symbol “–” indicates the following component is removed.

表 1 SEED 和 SEED-IV 上受试者相关分类准确率
(均值/标准差)

模型	SEED						SEED-IV
	delta band	theta band	alpha band	β波段	γ波段	所有波段	所有波段
SVM	60.50/14.14	60.95/10.20	66.64/14.41	80.76/11.56	79.56/11.38	83.99/09.92	56.61/20.05
GSCCA [59]	63.92/11.16	64.64/10.33	70.10/14.76	76.93/11.00	77.98/10.72	82.96/09.95	69.08/16.66
DBN [7]	64.32/12.45	60.77/10.42	64.01/15.97	78.92/12.48	79.19/14.58	86.08/08.34	66.77/07.38
STRNN [10]	80.90/12.27	83.35/09.15	82.69/12.99	83.41/10.16	69.61/15.65	89.50/07.63	DGCNN [12]
74.25/11.42	71.52/05.99	74.43/12.16	83.65/10.17	85.73/10.64	90.40/08.49	69.88/16.29	BiDANN [16]
76.97/10.95	75.56/07.88	81.03/11.74	89.65/09.59	88.64/09.46	92.38/07.04	70.29/12.63	EmotionMeter [24]
- 70.58/17.01	BiHDM [38]	(SOTA)	- 93.12/06.06	74.35/14.09	RGNN (Our model)	76.17/07.91	72.26/07.25
75.33/08.85	84.25/12.54	89.23/08.90	94.24/05.95	79.37/10.54			

表 2 在 SEED 和 SEED-IV 上的主题无关分类准确率
(均值/标准差)

模型	SEED						SEED-IV
	delta band	theta band	alpha band	β波段	γ波段	所有波段	所有波段
SVM	43.06/08.27	40.07/06.50	43.97/10.89	48.63/10.29	51.59/11.83	56.73/16.29	37.99/12.52
TCA [60]	44.10/08.22	41.26/09.21	42.93/14.33	43.93/10.06	48.43/09.73	63.64/14.88	56.56/13.77
SA [61]	53.23/07.47	50.60/08.31	55.06/10.60	56.72/10.78	64.47/14.96	69.00/10.89	64.44/09.46
T-SVM [62]	-	-	-	-	-	72.53/14.00	-
DGCNN [12]	49.79/10.94	46.36/12.06	48.29/12.28	56.15/14.01	54.87/17.53	79.95/09.02	52.82/09.23
DAN [63]	- - - -	-	-	-	-	-	83.81/08.56
BiDANN-S [16]	63.01/07.49	63.22/07.52	63.50/09.50	73.59/09.12	73.72/08.67	84.14/06.87	65.59/10.39
BiHDM [38] (SOTA)	- - - -	-	-	-	-	-	-
85.40/07.53	69.03/08.66	RGNN (我们的模型)	64.88/06.87	60.69/05.79	60.84/07.57	74.96/08.94	77.50/08.10
85.30/06.72	73.84/08.02						

改进可归因于两个因素：1) 我们的邻接矩阵包含了左右半球之间具有情绪区分性的全局跨通道不对称关系；2) 通过扩展 SGC，我们的模型减少了过拟合的担忧，而 SGC 比 DGCNN 中使用的 ChebNet[43] 简单得多。

6.2 对象无关分类

与表 1 和表 2 相似的是对象无关分类结果。当使用所有频段的特征时，我们的模型在 SEED 上的表现略差于 BiHDM，但在 SEED-IV 上表现明显优于 BiHDM（提高了近 5%）。此外

我们的模型在两个数据集上相对于所有基线都实现了最低的准确率标准差，显示了模型对跨被试变化的鲁棒性。

比较表 1 和表 2 中的结果，我们发现对于每个模型，在无被试设置下获得的准确率始终比在被试相关设置下获得的准确率低约 5%到 30%。这一发现并不令人意外，因为 EEG 信号在不同被试之间的变异性使得无被试分类更具挑战性。然而，一个有趣的观察是，这两种设置之间的性能差距正在逐渐减小，从使用 SVM 在 SEED 上约为 27%和在 SEED-IV 上约为 19%，到使用我们的模型在 SEED 上约为 9%和在 SEED-IV 上约为 6%。性能差距缩小的可能原因是，在无被试分类设置中的深度学习模型在利用大量数据和学习被试不变的 EEG 表示方面变得更好。这一观察似乎表明，迁移学习可能是跨被试设置中情绪识别的必要工具。

表 3 在 SEED 和 SEED-IV 上针对主体无关分类准确率（均值/标准差）的消融研究

模型	SEED	SEED-IV
RGNN	85.30/06.72	73.84/08.02
基于相关性的邻接矩阵	84.41/06.94	72.73/08.36
基于相干性的邻接矩阵	84.02/07.05	72.26/08.48
随机邻接矩阵	83.57/07.34	71.78/08.64
对称邻接矩阵	83.69/07.92	72.02/08.66
全局连接	82.42/08.24	71.13/08.78
全局连接替代方案 1	84.52/06.87	73.29/08.18
全局连接替代方案 2	84.23/07.04	73.08/08.35
NodeDAT	81.92/09.35	71.65/09.43
DAT	83.51/08.11	72.40/08.54
EmotionDL	82.27/08.81	70.76/09.22

6.3 频率带性能比较

我们在 SEED 上进一步比较了我们的模型和所有基线在不同频率带上的性能，如表 1 和表 2 所示。在受试者相关实验中，STRNN 在 delta、theta 和 alpha 带中达到最高准确率，BiDANN 在 beta 带中表现最佳，而我们的模型在 gamma 带中表现最佳。在受试者无关实验中，BiDANN-S 在 theta 和 alpha 带中达到最高准确率，而我们的模型在 delta、beta 和 gamma 带中表现最佳。



Fig. 3. Confusion matrices of RGNN. (a) Subject-dependent classification on SEED. (b) Subject-independent classification on SEED. (c) Subject-dependent classification on SEED-IV. (d) Subject-independent classification on SEED-IV.

We investigate the critical frequency bands for emotion recognition. For both subject-dependent and subject-independent settings on SEED, we compare the performance of each model across different frequency bands. In general, most models including ours achieve better performance on beta and gamma bands than delta, theta and alpha bands, with one exception of STRNN, which performs the worst on gamma band. This observation is consistent with the literature [7], [66]. One subtle difference between our model and other models is that our model performs consistently better in gamma band than beta band, whereas other models perform comparably in both bands, indicating that gamma band may be the most discriminative band for our model.

6.4 Confusion Matrix

We present the confusion matrices of our model in Fig. 3. For SEED, our model can recognize positive and neutral emotions better than negative emotion in both classification settings. Comparing subject-independent classification (see Fig. 3b) to subject-dependent classification (see Fig. 3a), the performance of our model gets relatively much worse at detecting negative emotion, indicating that participants are likely to generate distinct EEG patterns when experiencing negative emotion.

For SEED-IV, our model performs significantly better on sad emotion than all other emotions in both classification settings. Comparing subject-independent classification (see Fig. 3d) to subject-dependent classification (see Fig. 3c), the performance of our model gets relatively much worse at detecting sad emotion, which is similar to SEED. We note that fear is the only emotion that performs better in subject-independent classification than in subject-dependent classification. This finding indicates that participants watching horror movies may generate similar EEG patterns.

7 DISCUSSION

In this section, we conduct ablation study and sensitivity analysis for our RGNN model. We also analyze important brain regions and inter-channel relations for emotion recognition.

7.1 Ablation Study

We conduct ablation study to investigate the contribution of each key component in our model. Table 3 reports the subject-independent classification results on both datasets. We

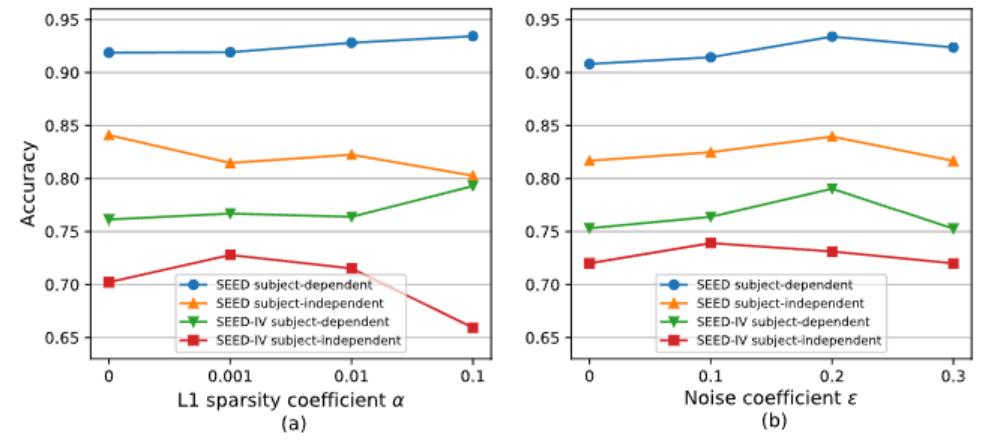


Fig. 4. Classification accuracy of RGNN with varying hyper-parameters. (a) L1 sparsity coefficient α in (11). (b) Noise coefficient ϵ in (15) and (16).

compared different initialization methods of the adjacency matrix and found that our distance-based method (see (9)) obtains slightly better performance than functional connectivity-based methods, i.e., correlation and coherence computed from the training dataset. The uniformly randomly initialized adjacency matrix in $[0, 1]$ performs worst, indicating that properly initializing the adjacency matrix is beneficial to model performance. Our symmetric adjacency matrix design also proves to be useful in reducing overfitting and improving accuracy.

Removing the global connection causes noticeable performance drop on both datasets, demonstrating the importance of global connections in modelling the EEG differential asymmetry. Moreover, we compared the performance of alternative sets of global connections. Alternative 1 has global indices that are nearer to the central region, i.e., (FP1, FP2), (AF3, AF4), (F3, F4), (FC3, FC4), (C3, C4), (CP3, CP4), (P3, P4), (PO5, PO6) and (O1, O2). Alternative 2 has global indices that are further from the central region, i.e., (FP1, FP2), (AF3, AF4), (F7, F8), (FT7, FT8), (T7, T8), (TP7, TP8), (P7, P8), (PO7, PO8) and (O1, O2). Both alternatives perform slightly worse than our model but much better than no global connection, indicating that they are able to model EEG asymmetry to a certain extent.

Our NodeDAT regularizer has a noticeable positive impact on the performance of our model, suggesting that domain adaptation is helpful in cross-subject classification. To further investigate the impact of our node-level domain classifier, we experimented with replacing NodeDAT with a generic domain classifier DAT [22]. The clear performance gap between DAT and our RGNN model indicates that NodeDAT can better regularize the model by learning subject-invariant representation at node level than graph level. In addition, if NodeDAT is removed, the performance of our model has a greater variance, validating the importance of our NodeDAT regularizer in improving the robustness of RGNN against cross-subject variations.

Our EmotionDL regularizer improves the performance of our model by around 3 percent in accuracy on both datasets. This performance gain validates our assumption that participants are not always generating the intended emotions when watching emotion-eliciting stimuli. In addition, our EmotionDL regularizer can be easily adopted by other deep learning based emotion recognition models.

7.2 Sensitivity Analysis

We analyze the performance of our model across varying L1 sparsity coefficient α (see (11)) and noise coefficient ϵ in EmotionDL (see (15) and (16)), as illustrated in Fig. 4. For

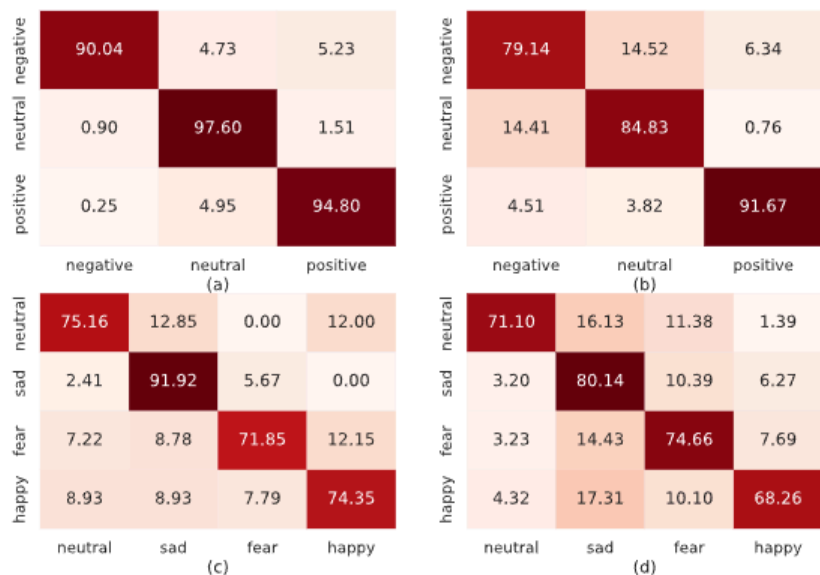


图 3. RGNN 的混淆矩阵。(a) SEED 上的受试者相关分类。(b) SEED 上的受试者无关分类。(c) SEED-IV 上的受试者相关分类。(d) SEED-IV 上的受试者无关分类。

我们研究了用于情绪识别的关键频段。在 SEED 的受试者相关和受试者无关设置下，我们比较了每个模型在不同频段上的性能。通常，大多数模型包括我们的模型在 beta 和 gamma 频段上的性能优于 delta、theta 和 alpha 频段，STRNN 除外，它在 gamma 频段上的性能最差。这一观察结果与文献[7]、[66]中的报道一致。我们模型与其他模型的一个微妙区别在于，我们的模型在 gamma 频段上的表现始终优于 beta 频段，而其他模型在这两个频段上的表现相当，这表明 gamma 频段可能是我们模型最具判别性的频段。

6.4 混淆矩阵

我们在图 3 中展示了我们模型的混淆矩阵。对于 SEED，我们的模型在两种分类设置中识别积极和中性情绪的能力都优于识别消极情绪。比较受试者无关分类（见图 3b）与受试者相关分类（见图 3a），我们的模型在检测消极情绪方面的性能相对恶化，这表明当参与者体验消极情绪时，他们可能会产生不同的脑电图模式。

对于 SEED-IV，我们的模型在两种分类设置中识别悲伤情绪的性能显著优于其他所有情绪。比较受试者无关分类（见图 3d）与受试者相关分类（见图 3c），我们的模型在检测悲伤情绪方面的性能相对恶化，这与 SEED 的情况相似。我们注意到，恐惧是唯一在受试者无关分类中表现优于受试者相关分类的情绪。这一发现表明，观看恐怖电影的参与者可能会产生相似的脑电图模式。

7 讨论

在本节中，我们对 RGNN 模型进行了消融研究和敏感性分析，并分析了用于情绪识别的重要脑区和通道间关系。

7.1 消融研究

我们进行消融研究以探究模型中每个关键组件的贡献。表 3 报告了在两个数据集上的受试者无关分类结果

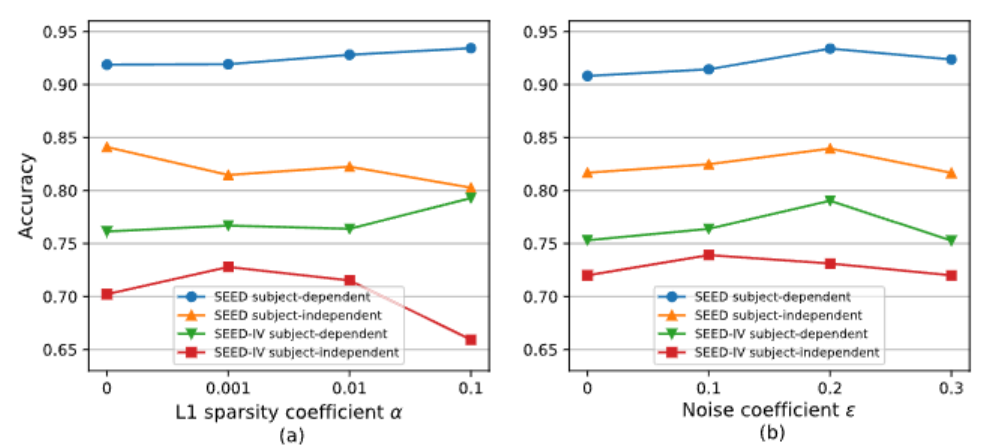


图 4. RGNN 的分类精度随超参数变化。(a) 公式(11)中的 L1 稀疏系数 α 。(b) 公式(15)和(16)中的噪声系数。

我们比较了邻接矩阵的不同初始化方法，发现基于距离的方法（见公式(9)）比基于功能连接的方法（即从训练数据集中计算的相关性和相干性）性能略好。在[0, 1]范围内均匀随机初始化的邻接矩阵表现最差，表明适当地初始化邻接矩阵有利于模型性能。我们的对称邻接矩阵设计也被证明有助于减少过拟合并提高准确性。

移除全局连接导致在两个数据集上的性能明显下降，这表明全局连接在建模脑电图差异不对称性中的重要性。此外，我们比较了替代全局连接集的性能。替代方案 1 的全局索引更接近中央区域，即 (FP1, FP2), (AF3, AF4), (F3, F4), (FC3, FC4), (C3, C4), (CP3, CP4), (P3, P4), (PO5, PO6)和(O1, O2)。替代方案 2 的全局索引更远离中央区域，即(FP1, FP2), (AF3, AF4), (F7, F8), (FT7, FT8), (T7, T8), (TP7, TP8), (P7, P8), (PO7, PO8)和(O1, O2)。这两种替代方案的性能略差于我们的模型，但远好于没有全局连接的情况，表明它们在一定程度上能够建模脑电图不对称性。

我们的 NodeDAT 正则化器对模型性能有显著的积极影响，表明领域自适应有助于跨主题分类。为了进一步研究我们节点级领域分类器的影响，我们尝试用通用的领域分类器 DAT [22] 替换 NodeDAT。DAT 和我们的 RGNN 模型之间的明显性能差距表明，NodeDAT 通过在节点级别而非图级别学习主题不变表示，能更好地正则化模型。此外，如果移除 NodeDAT，模型的性能具有更大的方差，验证了我们的 NodeDAT 正则化器在提高 RGNN 对跨主题变化鲁棒性方面的重要性。

我们的 EmotionDL 正则化器在两个数据集上使模型的准确率提高了约 3%。这一性能提升验证了我们的假设，即参与者在观看情绪诱发刺激时并不总是产生预期的情绪。此外，EmotionDL 正则化器可以很容易地被其他基于深度学习的情绪识别模型采用。

7.2 敏感性分析

我们分析了我们的模型在不同 L1 稀疏系数 α （见公式(11)）和 EmotionDL 中的噪声系数（见公式(15)和(16)）下的性能，如图 4 所示。对于

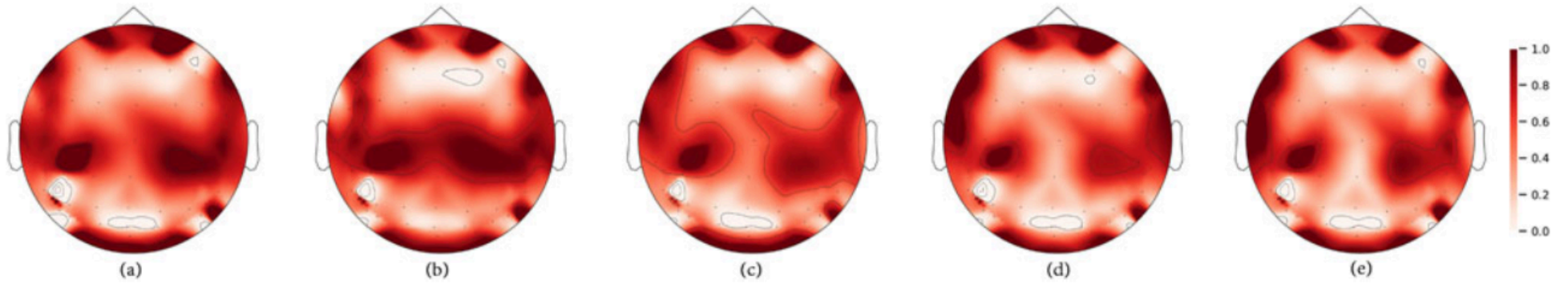


Fig. 5. Activation maps learned from subject-dependent classification on SEED-IV. (a) Delta band. (b) Theta band. (c) Alpha band. (d) Beta band. (e) Gamma band.

subject-dependent classification, increasing α from 0 to 0.1 generally increases the model performance. However, for subject-independent classification, increasing α beyond a certain threshold, i.e., 0.01 in Fig. 4a, decreases the model performance. One possible explanation for the difference in model behaviors is that there is much less training data in subject-dependent classification, which thus requires a stronger regularization to reduce overfitting, whereas for subject-independent classification where the amount of training data is less of a concern, adding stronger regularization may introduce bias and hinder the learning efficacy.

As illustrated in Fig. 4b, our model behaves consistently across different experimental settings with varying noise coefficient ϵ . Specifically, by increasing ϵ , the performance of our model first increases and then decreases. In particular, our model usually performs best when ϵ is set to 0.2, demonstrating the existence of label noises and the necessity of addressing them on both datasets. Introducing excessive noise in EmotionDL causes performance drop, which is expected because excessive noise weakens the true learning signals.

7.3 Analysis of Important Brain Regions and Inter-Channel Relations

We identify important brain regions for emotion recognition. Fig. 5 shows the heatmaps of the diagonal elements in our learned adjacency matrix $\mathbf{A}$ in subject-dependent classification on SEED-IV for each frequency band. The values are scaled to the $[0, 1]$ interval for better visualization. Conceptually, as shown in (4), the diagonal values in $\mathbf{A}$ represents the contribution of each channel in computing the final EEG representation. It is clear from 5 that there is strong activation on the pre-frontal, parietal and occipital regions for all frequency bands, indicating that these regions may be strongly related to the emotion processing in the brain. Our finding is consistent with existing studies, which observed that

asymmetrical frontal and parietal EEG activity may reflect changes on both valence and arousal [5], [27]. The synchronization between frontal and occipital regions has also been reported to be related to positive emotions [67]. In addition, there is strong activation on the temporal regions for beta and gamma bands, which is consistent with [7]. The symmetry pattern on the activation maps of channels also indicates that the asymmetry in EEG activity between the left and right hemispheres is critical for emotion recognition.

We identify important inter-channel relations for emotion recognition. Fig. 6 shows the top 10 connections between channels having the largest edge weights in our adjacency matrix $\mathbf{A}$. Note that all global connections remain among the strongest connections after $\mathbf{A}$ is learned, demonstrating again that global inter-channel relations are essential for emotion recognition. It is clear from Fig. 6b that the connection between the channel pair (FP1, AF3) is the strongest, followed by (F6, F8), (FP2, AF4) and (PO8, CB2), indicating that local inter-channel relations in the frontal region may be important for emotion recognition.

8 CONCLUSION

In this paper, we propose a regularized graph neural network for EEG-based emotion recognition. Our model is inspired by neuroscience theories on human brain organization and captures both local and global inter-channel relations in EEG signals. In addition, we propose two regularizers, namely NodeDAT and EmotionDL, to improve the robustness of our model against cross-subject EEG variations and noisy labels, respectively. Extensive experiments on two public datasets demonstrate the superior performance of our model than several competitive baselines and the state-of-the-art BiHDM in most experimental settings. Our model analysis shows that our proposed biologically inspired adjacency matrix and two regularizers contribute consistent and significant gain to the performance of our model. Investigations on the brain regions reveal that pre-frontal, parietal and occipital regions may be the most informative regions for emotion recognition. In addition, global inter-channel relations between the left and right hemispheres are important, and local inter-channel relations between (FP1, AF3), (F6, F8) and (FP2, AF4) may also provide useful information.

In the future, we plan to explore: 1) training a more discriminative domain classifier, e.g., by using more advanced classifiers or applying more sophisticated techniques to handle imbalanced samples between training and test sets, to help our model learn more domain-invariant EEG representations; 2) applying our model to EEG signals that have a smaller number of channels. A simpler version of our model and more

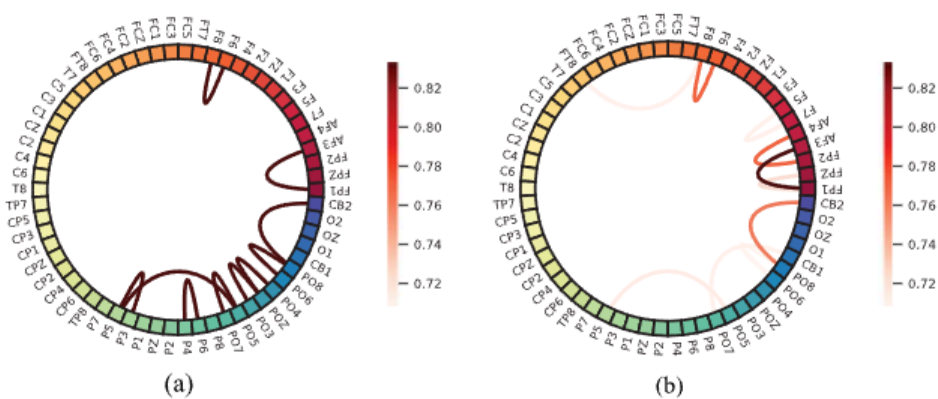


Fig. 6. Top 10 connections between channels in the adjacency matrix $\mathbf{A}$, excluding global connections in (10) for better clarity. (a) Initialized $\mathbf{A}$ according to (9). (b) Learned and averaged $\mathbf{A}$ across five frequency bands in subject-dependent classification on both SEED and SEED-IV.

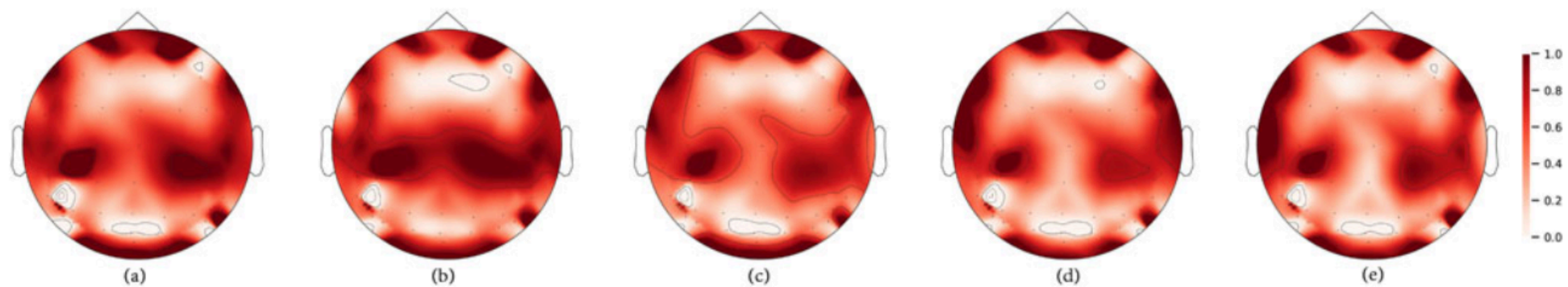


图 5. 从 SEED-IV 的受试者相关分类中学习到的激活图。(a) Delta 波段。(b) Theta 波段。(c) Alpha 波段。(d) Beta 波段。(e) Gamma 波段。

基于被试的分类，将 α 从 0 增加到 0.1 通常会提高模型性能。然而，对于基于被试的分类，当 α 超过某个阈值，即图 4a 中的 0.01 时，模型性能会下降。模型行为差异的一个可能解释是，基于被试的分类中训练数据较少，因此需要更强的正则化来减少过拟合，而对于基于被试的分类，训练数据的数量不是主要问题，增加更强的正则化可能会引入偏差并阻碍学习效率。

如图 4b 所示，我们的模型在不同实验设置下表现出一致性，噪声系数不断变化。具体来说，随着 α 的增加，我们模型的性能先提高后降低。特别地，当 α 设置为 0.2 时，我们的模型通常表现最佳，这表明标签噪声的存在以及处理这些噪声的必要性，在两个数据集上都得到了验证。在 EmotionDL 中引入过多的噪声会导致性能下降，这是可以预料的，因为过多的噪声会削弱真实的学习信号。

观察到不对称的前额叶和顶叶脑电图活动可能反映效价和唤醒度的变化[5], [27]。额叶和顶叶区域之间的同步性也与积极情绪有关[67]。此外，额叶区域的 Beta 和 Gamma 频段存在强烈激活，这与[7]一致。通道激活图上的对称模式也表明，左右半球脑电图活动中的不对称性对于情绪识别至关重要。

我们识别出对情绪识别重要的通道间关系。图 6 显示了在我们的邻接矩阵 A 中具有最大边重的通道间前 10 个连接。注意，在 A 学习后，所有全局连接仍然是最强的连接之一，这再次证明全局通道间关系对于情绪识别至关重要。从图 6b 可以看出，通道对(FP1, AF3)之间的连接是最强的，其次是(F6, F8), (FP2, AF4)和(PO8, CB2)，表明额叶区域的局部通道间关系可能对情绪识别很重要。

7.3 重要脑区分析及通道间关系分析

我们识别出对情绪识别重要的脑区。图 5 展示了在 SEED-IV 数据集上，针对每个频段，在受试者相关分类中，我们学习到的邻接矩阵 A 的对角线元素的热图。这些值被缩放到 $[0, 1]$ 区间，以便更好地可视化。从概念上讲，如图(4)所示， A 中的对角线值表示每个通道在计算最终脑电图表示时的贡献。从图 5 可以看出，所有频段的前额叶、顶叶和枕叶区域都有强烈的激活，这表明这些区域可能与大脑中的情绪处理密切相关。我们的发现与现有研究一致。

8 结论

在本文中，我们提出了一种用于基于脑电图（EEG）的情绪识别的正则化图神经网络。我们的模型受人类大脑组织神经科学理论的启发，能够捕捉脑电图信号中的局部和全局跨通道关系。此外，我们提出了两种正则化器，即 NodeDAT 和 EmotionDL，分别用于提高模型对跨被试 EEG 变化和噪声标签的鲁棒性。在两个公共数据集上的大量实验表明，我们的模型在大多数实验设置中优于几种竞争基线和最先进的 BiHDM，表现出优越的性能。模型分析显示，我们提出的受生物学启发的邻接矩阵和两种正则化器为模型性能带来了持续且显著的提升。对大脑区域的调查揭示，前额叶、顶叶和枕叶可能是情绪识别中最具信息量的区域。此外，左右半球之间的全局跨通道关系很重要，而 (FP1, AF3)、(F6, F8) 和 (FP2, AF4) 之间的局部跨通道关系也可能提供有用信息。

未来，我们计划探索：1) 训练一个更具判别性的领域分类器，例如通过使用更先进的分类器或应用更复杂的技术来处理训练集和测试集之间不平衡的样本，以帮助我们的模型学习更多领域不变的 EEG 表示；2) 将我们的模型应用于通道数更少的 EEG 信号。我们模型的简化版本和更多

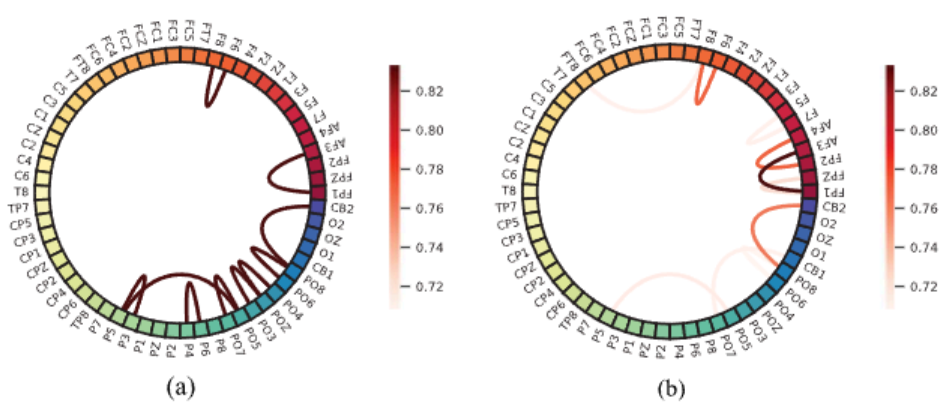


图 6. 在邻接矩阵 A 中，排除(10)中的全局连接以增强清晰度，显示通道间前 10 个连接。(a) 根据公式(9)初始化 A 。(b) 在 SEED 和 SEED-IV 的受试者相关分类中学习并平均 A ，涵盖五个频段。

advanced regularizations may be necessary to avoid over-smoothing on these small graphs. In addition, data processing techniques that can improve the spatial resolution of EEG signals, e.g., spatial filtering, may be worth exploring.

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为了防止在这些小图上出现过度平滑, 可能需要采用高级正则化方法。此外, 能够提高脑电图信号空间分辨率的数据处理技术, 例如空间滤波, 也可能值得探索。

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